Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**CALCULATION** **3110 PRIMARY CRUSHING** **GRADE SLAB**
Document Number:	**MD-216-3110-EG-CV-CAL-1005**
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
Employer	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of Primary Crusher area (3110)- HGTU tower foundation & others miscellaneous foundation. The member sizes and reinforcements are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD Pro. The STAAD modelling has been done with beam & plate elements. Plate mat with subgrade modulus has been considered as per L&T approved Soil Report. Base Pressure check has been done from STAAD Pro output.

For strength calculation Bending moment & Shear Force values are taken from STAAD output & designed in the excel sheet.ment-2.

4. DESIGN REQUIREMENTS

4.1. Codal References

Reference is made in this standard as indicated in Civil,Structural and Architectural Design Criteria Doc No. : MD-216-1000-EG-ST-CRT-1001-A02 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

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4.2. Materials Property

Density of Material
- Density of Dry Soil : 1.9 T/m^3
- Density of Concrete : 2.5 T/m^3
- Density of Water : 1.0 T/m^3

Concrete Strength of 28 days
- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade
- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

Grout Grade
- Grout shall be cementitious under steel column bases of Strength 40 MPa

4.3. Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	Minimum Cove (mm)
Exposed to Soil	Foundation Bottom & Side	75
	Foundation Top	50
	Column & Beam	50
Exposed to Weather	Column	50
	Beam	50
	Slab	40

4.5. Allowable Soil Bearing Capacity

Recommended net allowable bearing capacity as per L&T soil report MD-216-3000-EG-CV-RPT-1008-A02, bore hole BH-19.

BH-no	Depth of foundation below FGL(m)	Recommended net allowable bearing capacity(KN/m2)	Vertical modulus of subgrage reaction(KN/m2/m)
BH-19	1.5	600	72000

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5. SAFETY FACTOR AGAINST STABILITY

5.1. Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0

5.2. Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

6. LOAD CALCULATIONS

6.1. Primary Load Cases for HGTU tower foundation

1 DEAD LOAD (DL)
2 LIVE LOAD (LL)
3 EQUIPMENT LOAD (EQL)
4 WIND LOAD WL (+X)
5 WIND LOAD WL (-X)
6 WIND LOAD WL (+Z)
7 WIND LOAD WL (-Z)
8 SEISMIC LOAD SL (+X)
9 SEISMIC LOAD SL (-X)
10 SEISMIC LOAD SL (+Z)
11 SEISMIC LOAD SL (-Z)
12 TEMPERATURE LOAD (TL)

6.1.1 Load Combinations for Stress Calculation

LOAD CASE	LOAD COMBINATION
201	DL+LL+EQL+TL
202	DL+1.0(LL+EQL)+1.0TL
203	DL+WL(+X)
204	DL+WL(-X)
205	DL+WL(+Z)
206	DL+WL(-Z)
207	DL+0.7SL(+X)
208	DL+0.7SL(-X)
209	DL+0.7SL(+Z)
210	DL+0.7SL(-Z)
211	DL+0.75WL(+X)+0.75(LL+EQL)
212	DL+0.75WL(-X)+0.75(LL+EQL)
213	DL+0.75WL(+Z)+0.75(LL+EQL)
214	DL+0.75WL(-Z)+0.75(LL+EQL)
215	DL+0.525SL(+X)+0.75(LL+EQL)
216	DL+0.525SL(-X)+0.75(LL+EQL)
217	DL+0.525SL(+Z)+0.75(LL+EQL)
218	DL+0.525SL(-Z)+0.75(LL+EQL)
219	0.6(DL)+WL(+X)
220	0.6(DL)+WL(-X)
221	0.6(DL)+WL(+Z)
222	0.6(DL)+WL(-Z)
223	0.6(DL)+0.7SL(+X)
224	0.6(DL)+0.7SL(-X)
225	0.6(DL)+0.7SL(+Z)
226	0.6(DL)+0.7SL(-Z)

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6.1.2 Load Combinations for Strength Calculation

LOAD CASE	LOAD COMBINATION
101	1.4(DL)
102	1.4(DL)+1.7(LL+EQL)
103	1.4(DL+TL)+1.7(LL+EQL)
104	1.2(DL)+0.8WL(+X)
105	1.2(DL)+0.8WL(-X)
106	1.2(DL)+0.8WL(+Z)
107	1.2(DL)+0.8WL(-Z)
108	1.2(DL)+1.6WL(+X)+0.5(LL+EQL)
109	1.2(DL)+1.6WL(-X)+0.5(LL+EQL)
110	1.2(DL)+1.6WL(+Z)+0.5(LL+EQL)
111	1.2(DL)+1.6WL(-Z)+0.5(LL+EQL)
112	1.2(DL)+SL(+X)+0.5(LL+EQL)
113	1.2(DL)+SL(-X)+0.5(LL+EQL)
114	1.2(DL)+SL(+Z)+0.5(LL+EQL)
115	1.2(DL)+SL(-Z)+0.5(LL+EQL)
116	0.9(DL)+1.6WL(+X)
117	0.9(DL)+1.6WL(-X)
118	0.9(DL)+1.6WL(+Z)
119	0.9(DL)+1.6WL(-Z)
120	0.9(DL)+SL(+X)
121	0.9(DL)+SL(-X)
122	0.9(DL)+SL(+Z)
123	0.9(DL)+SL(-Z)

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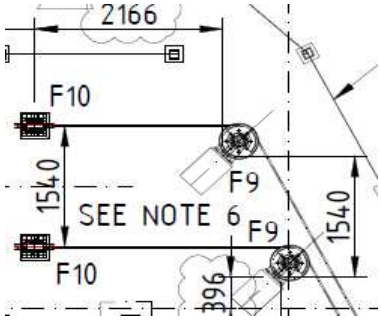
6.2. Primary Load Cases for F2 foundation

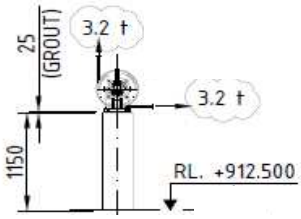
1 DEAD LOAD (DL)

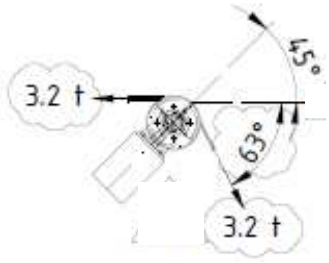
Dead load calculation :

a.Selfweight

b. Load from F9 & F10 support



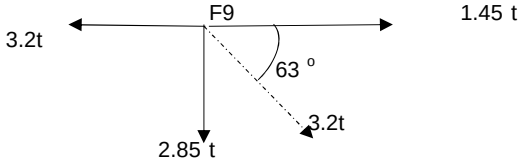




Load for F10 refer dwg no .MD-216-3110-EG-ME-LPX-1002-A03

Load for F9 refer dwg no. MD-216-3110-EG-ME-LPX-1008-A02

For load F9



c.Load from soil above footing

Depth = 0.6m unit weight of soil = 1.9t/m^3

Soil pressure = 0.6*1.9 = 1.14 t/m^2

6.2.1. Load Combinations for Stress Calculation

LOAD CASE	LOAD COMBINATION
101	DL

6.2.2 Load Combinations for Strength Calculation

LOAD CASE	LOAD COMBINATION
201	1.4DL

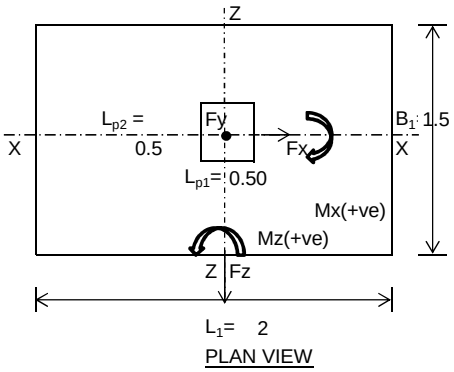
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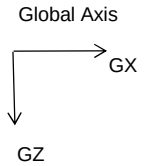
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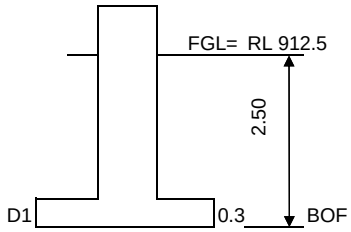
7.1 Design of Foundations

Foundation MKD. **F1**



PLAN VIEW





SECTIONAL VIEW

Foundation Level Details :

RL of Finish Ground Level ,FGL =	RL	912.5	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		600	kN/m^2

Material Properties :

Unit Weight of Soil =		19.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2	m
Footing dimension along Z direction,B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap =	0.5	x	0.5	=	0.25	m^2
Area of Footing , A =	2.00	x	1.50	=	3	m^2
Zz-z =	2	x	2.00 x 1.50 /	6	=	1	m^3
Zx-x =	1.5	x	1.5 x 2 /	6	=	0.75	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	22.5	kN
Self weight of Column =	-1.9	kN
Wt of soil above ftg =	115	kN
Total weight,W=	136	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.5	m
Pedestal dimension along Z direction,Lp2=	0.5	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 19 = 648 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F1 STAAD node100	Force from Load Combinations						Bearing Pressure Calculation						Check for Overturning:								Check for Sliding					
	Axial Force			Moment									About Z - Axis				About X - Axis									
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zzz+Mx/Zxx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zzz-Mx/Zxx	Remarks		Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2 + Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m²	kN/m²		kN/m²			kN-m	kN-m			kN-m	kN-m			kN	kN	kN	kN		
201	0.11	-198	0.63	1.88	0	-0.3	68.7	647.5	safe	63.1	ok		0.3	198	610	ok	1.9	148	78.7	ok	0.1	0.6	0.6	198	126	ok
202	0.11	-198	0.63	1.88	0	-0.3	68.7	647.5	safe	63.1	ok		0.3	198	610	ok	1.9	148	78.7	ok	0.1	0.6	0.6	198	126	ok
203	-0.5	-160	0.09	0.27	0	1.5	55.3	647.5	safe	51.6	ok		1.5	160	107	ok	0.3	120	454	ok	0.5	0.1	0.5	160		ok
204	0.5	-187	0.09	0.27	0	-1.5	64.1	647.5	safe	60.4	ok		1.5	187	124	ok	0.3	140	528	ok	0.5	0.1	0.5	187	849	ok
205	0	-168	-3.1	-9.2	0	0	68.3	647.5	safe	43.9	ok		0.0	168	-	-	9.2	126	13.8	ok	0.0	3.1	3.1	168	22	ok
206	0	-179	3.23	9.68	0	0	72.6	647.5	safe	46.8	ok		0.0	179	-	-	9.7	134	13.9	ok	0.0	3.2	3.2	179	22.2	ok
207	0	-165	0.09	0.27	0	0	55.5	647.5	safe	54.7	ok		0.0	165	-	-	0.3	124	467	ok	0.0	0.1	0.1	165	751	ok
208	0	-182	0.09	0.27	0	0	61.0	647.5	safe	60.3	ok		0.0	182	-	-	0.3	136	514	ok	0.0	0.1	0.1	182	826	ok
209	0	-170	-1	-3.1	0	0	61.0	647.5	safe	52.7	ok		0.0	170	-	-	3.1	128	41.1	ok	0.0	1.0	1.0	170	65.7	ok
210	0	-177	1.21	3.64	0	0	63.8	647.5	safe	54.1	ok		0.0	177	-	-	3.6	132	36.4	ok	0.0	1.2	1.2	177	58.2	ok
211	-0.37	-183	0.09	0.27	0	1.1	62.5	647.5	safe	59.6	ok		1.1	183	166	ok	0.3	137	518	ok	0.4	0.1	0.4	183	832	ok
212	0.38	-203	0.09	0.27	0	-1.1	69.2	647.5	safe	66.2	ok		1.1	203	177	ok	0.3	152	574	ok	0.4	0.1	0.4	203	922	ok
213	0.01	-189	-2.3	-6.8	0	-0	72.1	647.5	safe	53.9	ok		0.0	189	8587	ok	6.8	142	20.9	ok	0.0	2.3	2.3	189	33.4	ok
214	0.01	-197	2.44	7.33	0	-0	75.5	647.5	safe	55.9	ok		0.0	197	8956	ok	7.3	148	20.2	ok	0.0	2.4	2.4	197	32.3	ok
215	0.01	-187	0.09	0.27	0	-0	62.7	647.5	safe	61.9	ok		0.0	187	8488	ok	0.3	140	528	ok	0.0	0.1	0.1	187	849	ok
216	0.01	-199	0.09	0.27	0	-0	66.8	647.5	safe	66.1	ok		0.0	199	9055	ok	0.3	149	564	ok	0.0	0.1	0.1	199	905	ok
217	0.01	-191	-0.8	-2.3	0	-0	66.6	647.5	safe	60.5	ok		0.0	191	8665	ok	2.3	143	63	ok	0.0	0.8	0.8	191	101	ok
218	0.01	-195	0.93	2.8	0	-0	68.9	647.5	safe	61.4	ok		0.0	195	8877	ok	2.8	146	52.4	ok	0.0	0.9	0.9	195	83.7	ok
219	-0.5	-145	0.05	0.16	0	1.5	50.2	647.5	safe	46.8	ok		1.5	145	96.9	ok	0.2	109	685	ok	0.5	0.1	0.5	145	1096	ok
220	0.5	-172	0.05	0.16	0	-1.5	59.0	647.5	safe	55.6	ok		1.5	172	114	ok	0.2	129	810	ok	0.5	0.1	0.5	172	1296	ok
221	0	-153	-3.1	-9.3	0	0	63.4	647.5	safe	38.7	ok		0.0	153	-	-	9.3	115	12.4	ok	0.0	3.1	3.1	153	19.9	ok
222	0	-164	3.19	9.57	0	0	67.4	647.5	safe	41.9	ok		0.0	164	-	-	9.6	123	12.9	ok	0.0	3.2	3.2	164	20.6	ok
223	0	-150	0.05	0.16	0	0	50.3	647.5	safe	49.9	ok		0.0	150	-	-	0.2	113	708	ok	0.0	0.1	0.1	150	1133	ok
224	0	-167	0.05	0.16	0	0	55.9	647.5	safe	55.4	ok		0.0	167	-	-	0.2	125	787	ok	0.0	0.1	0.1	167	1259	ok
225	0	-155	-1.1	-3.2	0	0	56.1	647.5	safe	47.5	ok		0.0	155	-	-	3.2	117	36.2	ok	0.0	1.1	1.1	155	57.9	ok
226	0	-162	1.18	3.54	0	0	58.6	647.5	safe	49.2	ok		0.0	162	-	-	3.5	121	34.3	ok	0.0	1.2	1.2	162	54.8	ok

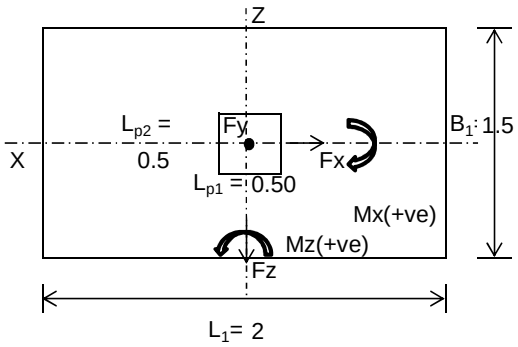
Design pressure for Strength Calculation

Foundation Mkd. F1 STAAD node100	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment						
	Fx	Fy	Fz	Mx	My	Mz	P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
L/C	kN	kN	kN	kN-m	KN-m	kN-m	kN/m^2			
101	0	-52	0.12	0.37	0	0	18.0	18	17.0	17
102	0.02	-97	0.12	0.37	0	-0.1	32.8	32.7	31.8	31.7
103	0.15	-94	0.88	2.64	0	-0.5	35.4	34.4	28.3	27.4
104	-0.4	-34	0.11	0.32	0	1.2	13.1	10.7	12.3	9.9
105	0.4	-56	0.11	0.32	0	-1.2	20.2	17.8	19.3	16.9
106	0	-41	-2.4	-7.2	0	0	23.2	23.2	4.0	4
107	0	-49	2.62	7.85	0	0	26.9	26.9	6.0	6
108	-0.8	-37	0.11	0.32	0	2.39	15.1	10.4	14.3	9.5
109	0.81	-79	0.11	0.32	0	-2.4	29.3	24.4	28.4	23.6
110	0.01	-49	-4.9	-15	0	-0	36.2	36.1	-3.3	-3.3
111	0.01	-67	5.13	15.4	0	-0	42.8	42.7	1.8	1.7
112	0.01	-46	0.11	0.32	0	-0	15.8	15.8	15.0	15
113	0.01	-70	0.11	0.32	0	-0	23.8	23.7	22.9	22.9
114	0.01	-71	-1.5	-4.5	0	-0	29.8	29.8	17.8	17.8
115	0.01	-62	1.71	5.14	0	-0	27.7	27.7	14.0	14
116	-0.8	-13	0.08	0.24	0	2.4	7.0	2.2	6.3	1.5
117	0.8	-55	0.08	0.24	0	-2.4	21.1	16.2	20.4	15.6
118	0	-25	-4.9	-15	0	0	28.2	28.2	-11.5	-11.5
119	0	-42	5.1	15.3	0	0	34.6	34.6	-6.3	-6.3
120	0	-22	0.08	0.24	0	0	7.6	7.6	7.0	7
121	0	-46	0.08	0.24	0	0	15.6	15.6	14.9	14.9
122	0	-29	-1.5	-4.6	0	0	15.9	15.9	3.7	3.7
123	0	-38	1.69	5.06	0	0	19.5	19.5	6.0	6

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PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	12	209.5	150	12	209.5	200

Footing Dimension :

Footing dimension along X direction, L_1	=	2	m
Footing dimension along Z direction, B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Column Dimension :

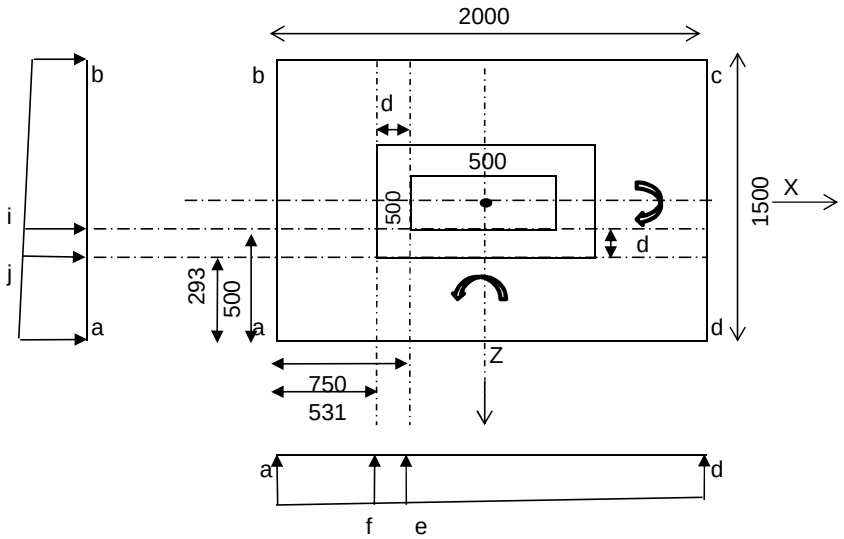
Pedestal dimension along X direction,Lp1	=	0.5	m
Pedestal dimension along Z direction,Lp2	=	0.5	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 42.8 kN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*42.8]*1.5*0.75*0.75	=	18.1	kN-m
Shear force at Face "f"	= 42.8*0.531*1.5	=	34.1	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*42.8]*2*0.5*0.5	=	10.7	kN-m
Shear force at Face "j"	= 42.8*0.293*2	=	25.1	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 2000 mm)

Cantilever Length			=	750	mm
Dia of Bar		d_b	=	12	mm
Effective Depth		d	=	219	mm
Bending Moment at "e"		M_u	=	18.1	kN-m
$C = 0.85f_c'ba$ and $T = A_sf_y$	Therefore 'a' = $f_y/0.85f_c'b * A_s$	a	=	0.0094	$*A_s$
$\varphi M_n \geq M_u = \varphi A_sf_y(d-a/2)$	0.9*420*(219-0.0094As/2)As=18.1*10^6	A_s	=	219.7	mm^2
P min	[SBC 304 Table 7.6.1.1]		=	0.18	%
$A_{s,min}$ =	0.18/100*1500*300	$A_{s,min}$	=	810	mm^2
Spacing required	1500*22/28*12^2/810	S_{reqd}	=	209.5	mm c/c
Spacing Provided		S_{prov}	=	150	mm c/c
A_s Provided	1500*22/28*12^2/150	$A_{s,pro}$	=	1131.5	mm^2
$a = A_sf_y/0.85f'_cb$	1131.5*420/(0.85*35*1500)	a	=	10.65	mm
$c = a/\beta_1$	10.65/0.8	c	=	13.32	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/13.32*(219-13.32)	ε_t	=	0.046	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u	=	34.1	kN
Moment at "f"	= [0.5*42.8]*1.5*0.531*0.531	M_u	=	9.1	kN-m
ρ_w provided	1131.5/1500/219	$\rho_{w,pro}$	=	0.0034	%
Effective Shear Area	=219*1500/10^6	bd	=	0.329	m^2
V_c = lesser of					
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$	=	326.8	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$	=	330.2	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$	=	563.6	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$			=	1207.1	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok				ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$	=	Not required	

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In Z Direction (Along Length 1500 mm)

Cantilever Length		=	500	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	207	mm
Bending Moment at "i"		M_u =	10.7	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0071	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(207-0.0071As/2)As=10.7*10^6	A_s =	137.1	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*2000*300	$A_{s,min}$ =	1080	mm^2
Spacing required	2000*22/28*12^2/1080	S_{reqd} =	209.52	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2000*22/28*12^2/200	$A_{s,pro}$ =	1131.5	mm^2
a = $A_{st} f_y$/0.85f'$_c$b	1131.5*420/(0.85*35*2000)	a =	7.99	mm
c = a/β_1	7.99/0.8	c =	9.99	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/9.99*(207-9.99)	ε_t =	0.059	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	25.1	kN
Moment at "j"	= [0.5*42.8]*2*0.293*0.293	M_u =	3.7	kN-m
ρ_w provided	= 1131.5/2000/207	$\rho_{w,pro}$ =	0.0027	%
Effective Shear Area	=207*2000/10^6	bd =	0.414	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$ =	419.0	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	411.1	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	710.3	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	1520.7	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

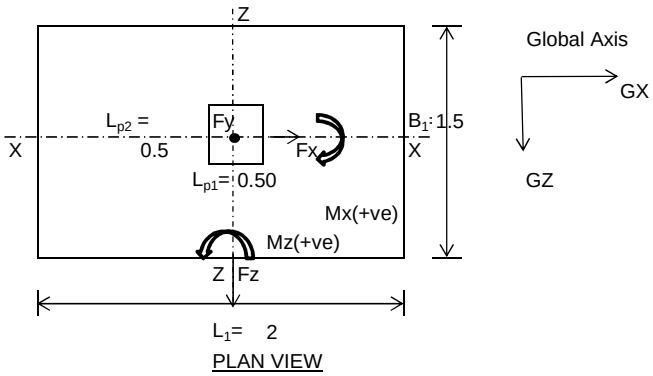
β = short side of column / long side of column = 1.00

Average soil pressure		=	42.80	kN/mm^2
Shear force,V_u	= 42.8*[2*1.5-(0.5+0.219)*(0.5+0.207)]	=	106.65	kN
Shear peremeter,b_0	= 2*[(0.5+0.219)+(0.5+0.207)]	=	2.85	m
$V_c = 0.17(1+2/\beta)\sqrt{f_c'}b_0d$	= 0.51 $\sqrt{f_c'}b_0d$			
$V_c = 0.083(\alpha_s d/b_0 + 2)\sqrt{f_c'}b_0d$	= 0.36 $\sqrt{f_c'}b_0d$	For Edge column	α_s	30
$V_c = 0.33\sqrt{f_c'}b_0d$	= 0.33 $\sqrt{f_c'}b_0d$			
φV_c	= 0.75*0.33*SQRT(35)*2852*219/1000	=	914.5	kN
Remarks : if $\varphi V_c > V_u$ ok	= safe			

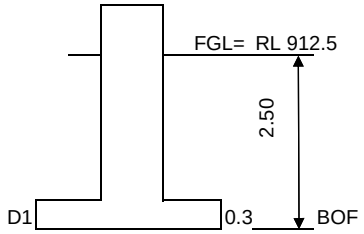
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Foundation MKD. **F1**



PLAN VIEW



SECTIONAL VIEW

Foundation Level Details :

RL of Finish Ground Level ,FGL =	RL	912.5	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		600	kN/m^2

Material Properties :

Unit Weight of Soil =		19.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2	m
Footing dimension along Z direction,B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap =	0.5	x	0.5	=	0.25	m^2
Area of Footing , A =	2.00	x	1.50	=	3	m^2
Zz-z =	2	x	2.00 x 1.50	/ 6	=	1	m^3
Zx-x =	1.5	x	1.5 x 2	/ 6	=	0.75	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	22.5	kN
Self weight of Column =	-1.88	kN
Wt of soil above ftg =	115	kN
Total weight,W=	136	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.5	m
Pedestal dimension along Z direction,Lp2=	0.5	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 19 = 648 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F1 STAAD node101	Force from Load Combinations						Bearing Pressure Calculation					Check for Overturning:									Check for Sliding					
	Axial Force			Moment								About Z - Axis				About X - Axis										
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zz+Mx/Zx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zz-Mx/Zx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2+Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	kN/m^2		kN/m^2		kN-m	kN-m			kN-m	kN-m			kN	kN	kN	kN		
201	19.01	-209	-0.6	-1.9	0	-57	129.4	647.5	safe	10.3	ok	57.0	209	3.68	ok	1.9	157	83.4	ok	19.0	0.6	19.0	209	133	ok
202	19.01	-209	-0.6	-1.9	0	-57	129.4	647.5	safe	10.3	ok	57.0	209	3.68	ok	1.9	157	83.4	ok	19.0	0.6	19.0	209	133	ok
203	-3.65	-149	-0.1	-0.3	0	10.9	61.0	647.5	safe	38.4	ok	10.9	149	13.6	ok	0.3	112	421	ok	3.6	0.1	3.6	149		ok
204	10.91	-199	-0.1	-0.3	0	-33	99.6	647.5	safe	33.4	ok	32.7	199	6.1	ok	0.3	150	564	ok	10.9	0.1	10.9	199	906	ok
205	2.697	-178	-3.2	-9.7	0	-8.1	80.3	647.5	safe	38.4	ok	8.1	178	22	ok	9.7	133	13.8	ok	2.7	3.2	4.2	178	22.1	ok
206	4.56	-170	3.04	9.12	0	-14	82.7	647.5	safe	31.0	ok	13.7	170	12.5	ok	9.1	128	14	ok	4.6	3.0	5.5	170	22.4	ok
207	0.457	-160	-0.1	-0.3	0	-1.4	55.0	647.5	safe	51.6	ok	1.4	160	116	ok	0.3	120	452	ok	0.5	0.1	0.5	160	726	ok
208	6.8	-189	-0.1	-0.3	0	-20	83.7	647.5	safe	42.2	ok	20.4	189	9.25	ok	0.3	141	534	ok	6.8	0.1	6.8	189	857	ok
209	3.683	-177	-1.2	-3.6	0	-11	75.0	647.5	safe	43.3	ok	11.1	177	16.1	ok	3.6	133	36.7	ok	3.7	1.2	3.9	177	58.8	ok
210	3.574	-171	1.03	3.09	0	-11	71.9	647.5	safe	42.2	ok	10.7	171	16	ok	3.1	128	41.5	ok	3.6	1.0	3.7	171	66.4	ok
211	5.18	-180	-0.1	-0.3	0	-16	76.0	647.5	safe	44.2	ok	15.5	180	11.6	ok	0.3	135	510	ok	5.2	0.1	5.2	180	819	ok
212	16.1	-218	-0.1	-0.3	0	-48	121.3	647.5	safe	24.1	ok	48.3	218	4.52	ok	0.3	163	617	ok	16.1	0.1	16.1	218	991	ok
213	9.939	-202	-2.4	-7.3	0	-30	106.9	647.5	safe	27.8	ok	29.8	202	6.77	ok	7.3	151	20.7	ok	9.9	2.4	10.2	202	33.2	ok
214	11.34	-196	2.26	6.77	0	-34	108.5	647.5	safe	22.4	ok	34.0	196	5.78	ok	6.8	147	21.7	ok	11.3	2.3	11.6	196	34.8	ok
215	8.259	-188	-0.1	-0.3	0	-25	87.9	647.5	safe	37.6	ok	24.8	188	7.6	ok	0.3	141	533	ok	8.3	0.1	8.3	188	855	ok
216	13.02	-210	-0.1	-0.3	0	-39	109.4	647.5	safe	30.6	ok	39.0	210	5.38	ok	0.3	157	594	ok	13.0	0.1	13.0	210	954	ok
217	10.68	-201	-0.9	-2.8	0	-32	102.9	647.5	safe	31.4	ok	32.0	201	6.29	ok	2.8	151	54.3	ok	10.7	0.9	10.7	201	86.9	ok
218	10.6	-197	0.75	2.25	0	-32	100.4	647.5	safe	30.8	ok	31.8	197	6.19	ok	2.3	148	65.5	ok	10.6	0.8	10.6	197	105	ok
219	-5.1	-134	-0.1	-0.2	0	15.3	60.1	647.5	safe	29.1	ok	15.3	134	8.74	ok	0.2	100	631	ok	5.1	0.1	5.1	134	1009	ok
220	9.454	-184	-0.1	-0.2	0	-28	90.0	647.5	safe	32.8	ok	28.4	184	6.5	ok	0.2	138	868	ok	9.5	0.1	9.5	184	1390	ok
221	1.245	-163	-3.2	-9.5	0	-3.7	70.7	647.5	safe	37.8	ok	3.7	163	43.5	ok	9.5	122	12.8	ok	1.2	3.2	3.4	163	20.5	ok
222	3.109	-155	3.08	9.23	0	-9.3	73.4	647.5	safe	30.1	ok	9.3	155	16.6	ok	9.2	116	12.6	ok	3.1	3.1	4.4	155	20.2	ok
223	-0.99	-144	-0.1	-0.2	0	2.98	51.4	647.5	safe	45.0	ok	3.0	144	48.4	ok	0.2	108	681	ok	1.0	0.1	1.0	144	1090	ok
224	5.349	-173	-0.1	-0.2	0	-16	74.1	647.5	safe	41.6	ok	16.0	173	10.8	ok	0.2	130	818	ok	5.3	0.1	5.3	173	1308	ok
225	2.232	-162	-1.2	-3.5	0	-6.7	65.4	647.5	safe	42.7	ok	6.7	162	24.2	ok	3.5	122	34.6	ok	2.2	1.2	2.5	162	55.3	ok
226	2.122	-156	1.07	3.2	0	-6.4	62.6	647.5	safe	41.3	ok	6.4	156	24.5	ok	3.2	117	36.5	ok	2.1	1.1	2.4	156	58.5	ok

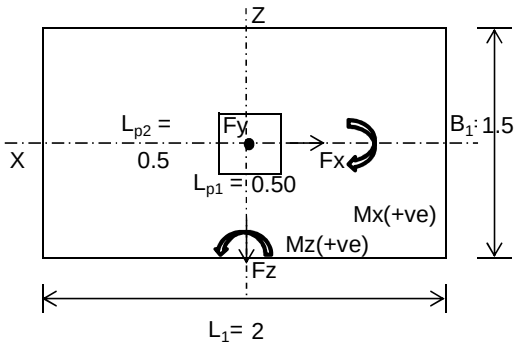
Design pressure for Strength Calculation

Foundation Mkd. F1 STAAD node101	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment			P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
	Fx	Fy	Fz	Mx	My	Mz				
L/C	kN	kN	kN	kN-m	KN-m	kN-m	kN/m^2			
101	5.08	-53	-0.1	-0.4	0	-15	33.6	3.1	32.6	2.1
102	20.97	-110	-0.1	-0.4	0	-63	100.0	-25.9	99.1	-26.8
103	29.41	-113	-0.9	-2.6	0	-88	129.4	-47.2	122.3	-54.2
104	-1.47	-26	-0.1	-0.3	0	4.4	13.4	4.6	12.6	3.8
105	10.18	-66	-0.1	-0.3	0	-31	53.0	-8.2	52.1	-9
106	3.609	-49	-2.6	-7.8	0	-11	37.6	15.9	16.7	-5.1
107	5.099	-43	2.4	7.19	0	-15	39.2	8.6	20.0	-10.7
108	-2.62	-22	-0.1	-0.3	0	7.85	15.7	-0.1	14.8	-1
109	20.67	-103	-0.1	-0.3	0	-62	96.7	-27.4	95.9	-28.2
110	7.536	-68	-5.1	-15	0	-23	65.9	20.7	25.0	-20.3
111	10.52	-56	4.9	14.7	0	-32	70.0	6.9	30.8	-32.4
112	4.496	-42	-0.1	-0.3	0	-13	27.9	0.9	27.0	-0.1
113	13.56	-83	-0.1	-0.3	0	-41	68.8	-12.6	68.0	-13.5
114	13.7	-82	-1.7	-5.1	0	-41	75.2	-7.1	61.6	-20.7
115	8.949	-58	1.49	4.48	0	-27	52.2	-1.6	40.2	-13.6
116	-8.38	6	-0.1	-0.2	0	25.1	23.5	-26.9	22.9	-27.5
117	14.91	-75	-0.1	-0.2	0	-45	70.0	-19.6	69.4	-20.2
118	1.775	-40	-5.1	-15	0	-5.3	39.1	28.5	-1.6	-12.3
119	4.756	-28	4.93	14.8	0	-14	43.5	14.9	4.1	-24.6
120	-1.27	-14	-0.1	-0.2	0	3.8	8.7	1.1	8.1	0.5
121	7.796	-55	-0.1	-0.2	0	-23	42.1	-4.8	41.5	-5.4
122	3.344	-39	-1.7	-5	0	-10	29.7	9.7	16.3	-3.9
123	3.187	-30	1.52	4.56	0	-9.6	25.6	6.5	13.5	-5.7

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PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	12	209.5	150	12	209.5	200

Footing Dimension :

Footing dimension along X direction, L_1	=	2	m
Footing dimension along Z direction, B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Column Dimension :

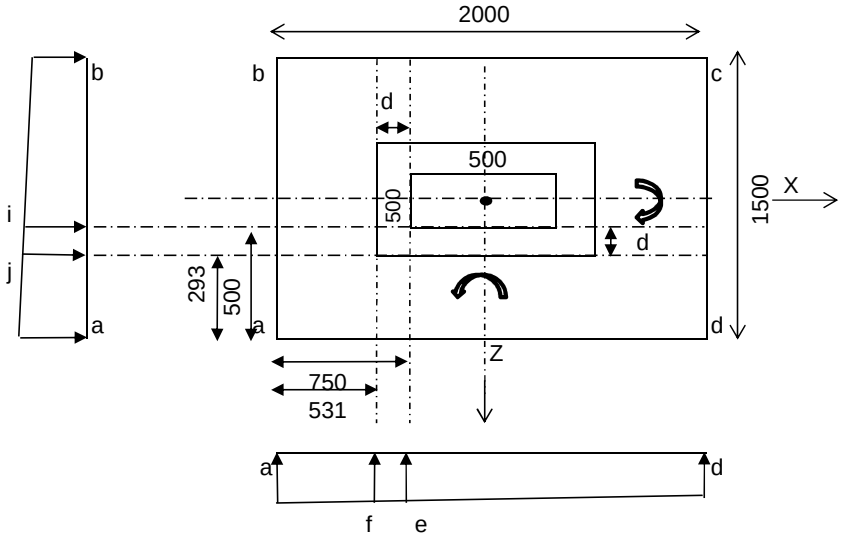
Pedestal dimension along X direction,Lp1	=	0.5	m
Pedestal dimension along Z direction,Lp2	=	0.5	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 129.4 kN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*129.4]*1.5*0.75*0.75	=	54.6	kN-m
Shear force at Face "f"	= 129.4*0.531*1.5	=	103.1	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*129.4]*2*0.5*0.5	=	32.4	kN-m
Shear force at Face "j"	= 129.4*0.293*2	=	75.8	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 2000 mm)

Cantilever Length		=	750	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	219	mm
Bending Moment at "e"		M_u =	54.6	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0094	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y (d-a/2)$	0.9*420*(219-0.0094As/2)As=54.6*10^6	A_s =	669.2	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*1500*300	$A_{s,min}$ =	810	mm^2
Spacing required	1500*22/28*12^2/810	S_{reqd} =	209.5	mm c/c
Spacing Provided		S_{prov} =	150	mm c/c
A_s Provided	1500*22/28*12^2/150	$A_{s,pro}$ =	1131.5	mm^2
a = $A_s f_y$/0.85f'$_c$b	1131.5*420/(0.85*35*1500)	a =	10.65	mm
c = a/β_1	10.65/0.8	c =	13.32	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/13.32*(219-13.32)	ε_t =	0.046	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	103.1	kN
Moment at "f"	= [0.5*129.4]*1.5*0.531*0.531	M_u =	27.4	kN-m
ρ_w provided	1131.5/1500/219	$\rho_{w,pro}$ =	0.0034	%
Effective Shear Area	=219*1500/10^6	bd =	0.329	m^2

V_c = lesser of

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$ =	326.8	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	330.2	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	563.6	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	1207.1	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 1500 mm)

Cantilever Length		=	500	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	207	mm
Bending Moment at "i"		M_u =	32.4	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0071	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(207-0.0071As/2)As=32.35*10^6	A_s =	416.4	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*2000*300	$A_{s,min}$ =	1080	mm^2
Spacing required	2000*22/28*12^2/1080	S_{reqd} =	209.52	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2000*22/28*12^2/200	$A_{s,pro}$ =	1131.5	mm^2
a = $A_{st} f_y$/0.85f'$_c$b	1131.5*420/(0.85*35*2000)	a =	7.99	mm
c = a/β_1	7.99/0.8	c =	9.99	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/9.99*(207-9.99)	ε_t =	0.059	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	75.8	kN
Moment at "j"	= [0.5*129.4]*2*0.293*0.293	M_u =	11.1	kN-m
ρ_w provided	= 1131.5/2000/207	$\rho_{w,pro}$ =	0.0027	%
Effective Shear Area	=207*2000/10^6	bd =	0.414	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$ =	419.1	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	411.1	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	710.3	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	1520.7	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	129.40	kN/mm^2
Shear force,V_u	= 129.4*[2*1.5-(0.5+0.219)*(0.5+0.207)]	=	322.43	kN
Shear peremeter,b_0	= 2*[(0.5+0.219)+(0.5+0.207)]	=	2.85	m

$V_c = 0.17(1+2/\beta)\sqrt{f_c'}b_0d$ = 0.51 $\sqrt{f_c'}b_0d$

$V_c = 0.083(\alpha_s d/b_0 + 2)\sqrt{f_c'}b_0d$ = 0.36 $\sqrt{f_c'}b_0d$

For Edge column	α_s	30

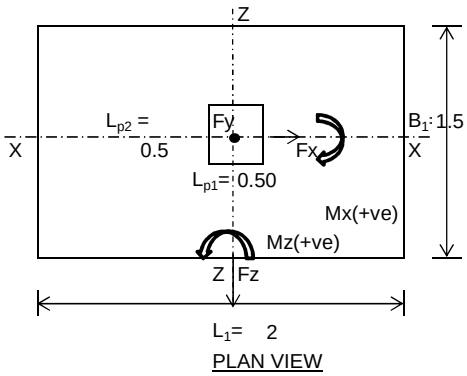
$V_c = 0.33\sqrt{f_c'}b_0d$ = 0.33 $\sqrt{f_c'}b_0d$

φV_c	= 0.75*0.33*SQRT(35)*2852*219/1000	=	914.5	kN
Remarks : if $\varphi V_c > V_u$ ok	= safe			

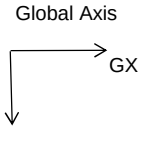
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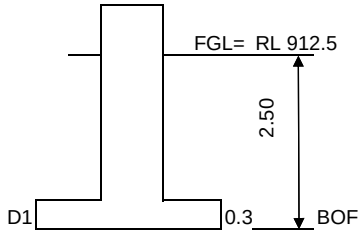
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Foundation MKD. **F1**



PLAN VIEW





SECTIONAL VIEW

Foundation Level Details :

RL of Finish Ground Level ,FGL =	RL	912.5	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		600	kN/m^2

Material Properties :

Unit Weight of Soil =		19.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2	m
Footing dimension along Z direction,B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap =	0.5	x	0.5	=	0.25	m^2
Area of Footing , A =	2.00	x	1.50	=	3	m^2
Zz-z =	2	x	2.00 x 1.50	/ 6	=	1	m^3
Zx-x =	1.5	x	1.5 x 2	/ 6	=	0.75	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	22.5	kN
Self weight of Column =	-1.88	kN
Wt of soil above ftg =	115	kN
Total weight,W=	136	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.5	m
Pedestal dimension along Z direction,Lp2=	0.5	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 19 = 648 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F1 STAAD node102	Force from Load Combinations						Bearing Pressure Calculation					Check for Overturning:								Check for Sliding					
	Axial Force			Moment								About Z - Axis				About X - Axis									
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zz+Mx/Zx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zz-Mx/Zx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2+Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	kN/m^2		kN/m^2		kN-m	kN-m			kN-m	kN-m			kN	kN	kN	kN		
201	24.81	-264	1.33	4	0	-74	168.0	647.5	safe	8.4	ok	74.4	264	3.56	ok	4.0	198	49.6	ok	24.8	1.3	24.8	264	79.3	ok
202	24.81	-264	1.33	4	0	-74	168.0	647.5	safe	8.4	ok	74.4	264	3.56	ok	4.0	198	49.6	ok	24.8	1.3	24.8	264	79.3	ok
203	-4.03	-183	0.14	0.41	0	12.1	73.6	647.5	safe	48.4	ok	12.1	183	15.1	ok	0.4	137	333	ok	4.0	0.1	4.0	183		ok
204	7.581	-179	0.16	0.47	0	-23	83.0	647.5	safe	36.2	ok	22.7	179	7.86	ok	0.5	134	284	ok	7.6	0.2	7.6	179	455	ok
205	1.648	-169	-5	-15	0	-4.9	81.3	647.5	safe	31.8	ok	4.9	169	34.3	ok	14.9	127	8.55	ok	1.6	5.0	5.2	169	13.7	ok
206	1.902	-192	5.25	15.7	0	-5.7	90.8	647.5	safe	37.4	ok	5.7	192	33.7	ok	15.7	144	9.16	ok	1.9	5.2	5.6	192	14.7	ok
207	-0.55	-184	0.14	0.42	0	1.64	63.5	647.5	safe	59.1	ok	1.6	184	112	ok	0.4	138	327	ok	0.5	0.1	0.6	184	525	ok
208	4.095	-178	0.15	0.46	0	-12	72.2	647.5	safe	46.4	ok	12.3	178	14.5	ok	0.5	133	288	ok	4.1	0.2	4.1	178	462	ok
209	1.061	-174	-2.5	-7.5	0	-3.2	71.2	647.5	safe	44.8	ok	3.2	174	54.6	ok	7.5	130	17.4	ok	1.1	2.5	2.7	174	27.8	ok
210	2.489	-188	2.8	8.39	0	-7.5	81.3	647.5	safe	43.9	ok	7.5	188	25.1	ok	8.4	141	16.8	ok	2.5	2.8	3.7	188	26.8	ok
211	2.886	-246	0.84	2.52	0	-8.7	93.9	647.5	safe	69.9	ok	8.7	246	28.4	ok	2.5	184	73.2	ok	2.9	0.8	3.0	246	117	ok
212	11.59	-242	0.85	2.56	0	-35	119.0	647.5	safe	42.6	ok	34.8	242	6.97	ok	2.6	182	71	ok	11.6	0.9	11.6	242	114	ok
213	7.144	-235	-3	-8.9	0	-21	111.9	647.5	safe	45.2	ok	21.4	235	11	ok	8.9	177	19.8	ok	7.1	3.0	7.7	235	31.6	ok
214	7.335	-252	4.67	14	0	-22	124.9	647.5	safe	43.5	ok	22.0	252	11.5	ok	14.0	189	13.5	ok	7.3	4.7	8.7	252	21.6	ok
215	5.5	-246	0.84	2.52	0	-16	102.0	647.5	safe	62.3	ok	16.5	246	14.9	ok	2.5	185	73.2	ok	5.5	0.8	5.6	246	117	ok
216	8.98	-242	0.85	2.55	0	-27	110.9	647.5	safe	50.3	ok	26.9	242	8.98	ok	2.6	181	71	ok	9.0	0.9	9.0	242	114	ok
217	6.704	-239	-1.1	-3.4	0	-20	104.3	647.5	safe	55.0	ok	20.1	239	11.9	ok	3.4	179	52.3	ok	6.7	1.1	6.8	239	83.7	ok
218	7.775	-249	2.83	8.5	0	-23	117.8	647.5	safe	48.4	ok	23.3	249	10.7	ok	8.5	187	22	ok	7.8	2.8	8.3	249	35.2	ok
219	-4.74	-165	0.08	0.24	0	14.2	69.6	647.5	safe	40.5	ok	14.2	165	11.6	ok	0.2	124	526	ok	4.7	0.1	4.7	165	846	ok
220	6.871	-161	0.1	0.29	0	-21	74.6	647.5	safe	32.6	ok	20.6	161	7.8	ok	0.3	121	410	ok	6.9	0.1	6.9	161	656	ok
221	0.938	-151	-5	-15	0	-2.8	73.4	647.5	safe	27.7	ok	2.8	151	53.9	ok	15.0	114	7.56	ok	0.9	5.0	5.1	151	12.1	ok
222	1.192	-174	5.19	15.6	0	-3.6	82.4	647.5	safe	33.8	ok	3.6	174	48.7	ok	15.6	131	8.4	ok	1.2	5.2	5.3	174	13.4	ok
223	-1.26	-166	0.08	0.24	0	3.77	59.4	647.5	safe	51.3	ok	3.8	166	44.1	ok	0.2	124	510	ok	1.3	0.1	1.3	166	819	ok
224	3.385	-160	0.1	0.29	0	-10	63.8	647.5	safe	42.8	ok	10.2	160	15.7	ok	0.3	120	421	ok	3.4	0.1	3.4	160	673	ok
225	0.351	-156	-2.6	-7.7	0	-1.1	63.3	647.5	safe	40.7	ok	1.1	156	148	ok	7.7	117	15.2	ok	0.4	2.6	2.6	156	24.4	ok
226	1.779	-170	2.74	8.21	0	-5.3	72.9	647.5	safe	40.3	ok	5.3	170	31.8	ok	8.2	127	15.5	ok	1.8	2.7	3.3	170	24.8	ok

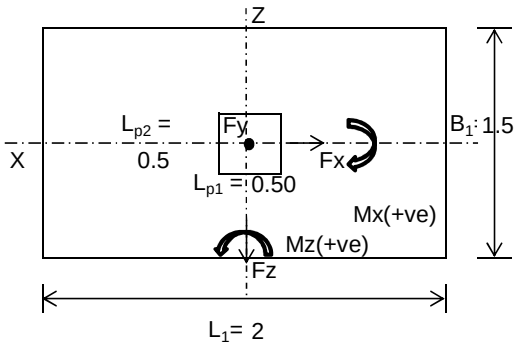
Design pressure for Strength Calculation

Foundation Mkd. F1 STAAD node102	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment			P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
	Fx	Fy	Fz	Mx	My	Mz				
L/C	kN	kN	kN	kN-m	KN-m	kN-m	kN/m^2			
101	2.485	-63	0.21	0.62	0	-7.5	29.2	14.3	27.6	12.6
102	14.87	-206	1.79	5.37	0	-45	120.5	31.2	106.1	16.9
103	36.92	-205	2.15	6.44	0	-111	187.8	-33.9	170.6	-51
104	-2.51	-55	0.17	0.51	0	7.54	26.7	11.6	25.4	10.3
105	6.774	-52	0.18	0.55	0	-20	38.4	-2.3	37.0	-3.8
106	2.028	-45	-3.9	-12	0	-6.1	36.6	24.4	5.4	-6.9
107	2.232	-63	4.26	12.8	0	-6.7	44.7	31.3	10.6	-2.8
108	-3.52	-99	0.63	1.88	0	10.5	46.2	25.1	41.2	20.1
109	15.06	-92	0.66	1.97	0	-45	78.7	-11.8	73.4	-17.1
110	5.569	-78	-7.5	-23	0	-17	72.7	39.3	12.6	-20.9
111	5.977	-114	8.8	26.4	0	-18	91.2	55.3	20.8	-15.2
112	2.459	-100	0.63	1.9	0	-7.4	43.3	28.6	38.3	23.5
113	9.088	-91	0.65	1.96	0	-27	60.4	5.9	55.2	0.7
114	6.494	-120	-2.6	-7.8	0	-19	70.0	31	49.2	10.3
115	6.793	-106	4.43	13.3	0	-20	73.4	32.6	38.0	-2.9
116	-7.69	-44	0.12	0.35	0	23.1	38.1	-8.1	37.2	-9
117	10.89	-37	0.15	0.44	0	-33	45.6	-19.8	44.4	-21
118	1.394	-22	-8	-24	0	-4.2	43.7	35.3	-20.6	-29
119	1.801	-58	8.29	24.9	0	-5.4	58.1	47.3	-8.3	-19.1
120	-1.72	-45	0.12	0.37	0	5.15	20.6	10.3	19.6	9.3
121	4.912	-36	0.14	0.43	0	-15	27.3	-2.2	26.2	-3.4
122	0.578	-30	-3.7	-11	0	-1.7	26.5	23.1	-2.8	-6.3
123	2.617	-50	3.92	11.8	0	-7.9	40.3	24.6	8.9	-6.9

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PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	12	173.7	150	12	209.5	200

Footing Dimension :

Footing dimension along X direction, L_1	=	2	m
Footing dimension along Z direction, B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Column Dimension :

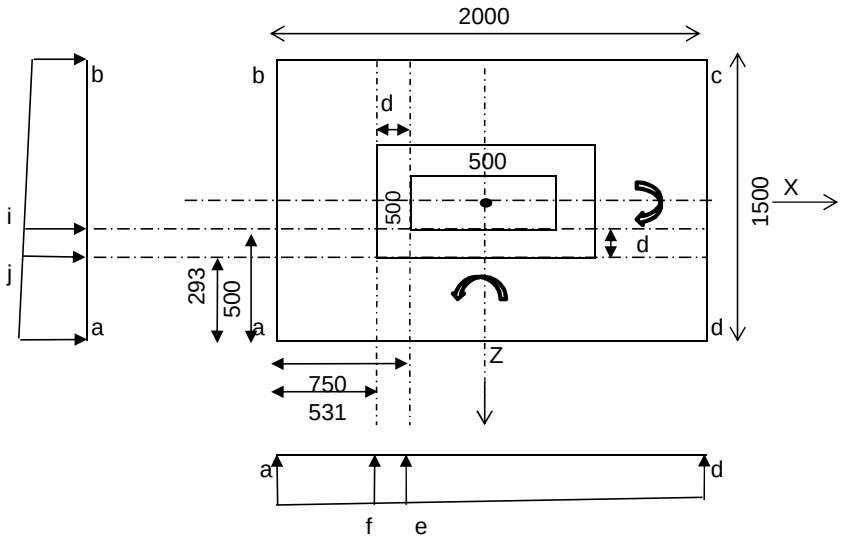
Pedestal dimension along X direction,Lp1	=	0.5	m
Pedestal dimension along Z direction,Lp2	=	0.5	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 187.8 kN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*187.8]*1.5*0.75*0.75	=	79.2	kN-m
Shear force at Face "f"	= 187.8*0.531*1.5	=	149.6	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*187.8]*2*0.5*0.5	=	47.0	kN-m
Shear force at Face "j"	= 187.8*0.293*2	=	110.1	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 2000 mm)

Cantilever Length		=	750	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	219	mm
Bending Moment at "e"		M_u =	79.2	kN-m
C = $0.85f_c'ba$ and T = A_sf_y	Therefore 'a' = $f_y/0.85f_c'b$ * A_s	a =	0.0094	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(219-0.0094As/2)As=79.2*10^6	A_s =	977.2	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*1500*300	$A_{s,min}$ =	810	mm^2
Spacing required	1500*22/28*12^2/977.2	S_{reqd} =	173.7	mm c/c
Spacing Provided		S_{prov} =	150	mm c/c
A_s Provided	1500*22/28*12^2/150	$A_{s,pro}$ =	1131.5	mm^2
$a = A_sf_y/0.85f'_cb$	1131.5*420/(0.85*35*1500)	a =	10.65	mm
$c = a/\beta_1$	10.65/0.8	c =	13.32	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/13.32*(219-13.32)	ε_t =	0.046	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	149.6	kN
Moment at "f"	= [0.5*187.8]*1.5*0.531*0.531	M_u =	39.7	kN-m
ρ_w provided	1131.5/1500/219	$\rho_{w,pro}$ =	0.0034	%
Effective Shear Area	=219*1500/10^6	bd =	0.329	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$ =	326.8	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	330.2	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	563.6	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	1207.1	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 1500 mm)

Cantilever Length		=	500	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	207	mm
Bending Moment at "i"		M_u =	47.0	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0071	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(207-0.0071As/2)As=46.95*10^6	A_s =	606.3	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*2000*300	$A_{s,min}$ =	1080	mm^2
Spacing required	2000*22/28*12^2/1080	S_{reqd} =	209.52	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2000*22/28*12^2/200	$A_{s,pro}$ =	1131.5	mm^2
a = $A_{st} f_y$/0.85f'$_c$b	1131.5*420/(0.85*35*2000)	a =	7.99	mm
c = a/β_1	7.99/0.8	c =	9.99	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/9.99*(207-9.99)	ε_t =	0.059	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	110.1	kN
Moment at "j"	= [0.5*187.8]*2*0.293*0.293	M_u =	16.1	kN-m
ρ_w provided	= 1131.5/2000/207	$\rho_{w,pro}$ =	0.0027	%
Effective Shear Area	=207*2000/10^6	bd =	0.414	m^2
V_c = lesser of				
a) (0.16λ$\sqrt{f_c'}$ + 17$\rho_w V_u d/M_u$)bd		$V_{c,a}$ =	419.0	kN
b) (0.16λ$\sqrt{f_c'}$ + 17ρ_w)bd		$V_{c,b}$ =	411.1	kN
c) 0.29λ$\sqrt{f_c'}$bd		$V_{c,c}$ =	710.3	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	1520.7	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	187.80	kN/mm^2
Shear force,V_u	= 187.8*[2*1.5-(0.5+0.219)*(0.5+0.207)]	=	467.94	kN
Shear peremeter,b_0	= 2*[(0.5+0.219)+(0.5+0.207)]	=	2.85	m

V_c =0.17(1+2/β)$\sqrt{f_c'}b_0d$ = 0.51 $\sqrt{f_c'}b_0d$

$V_c = 0.083(\alpha_s d/b_0 + 2)\sqrt{f_c'}b_0d$ = 0.36 $\sqrt{f_c'}b_0d$ — For Edge column, α_s = 30

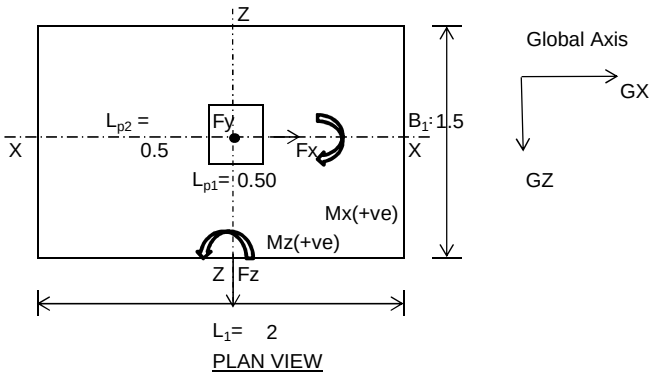
V_c = 0.33$\sqrt{f_c'}b_0d$ = 0.33 $\sqrt{f_c'}b_0d$

φV_c	= 0.75*0.33*SQRT(35)*2852*219/1000	=	914.5	kN
Remarks : if $\varphi V_c > V_u$ ok	= safe			

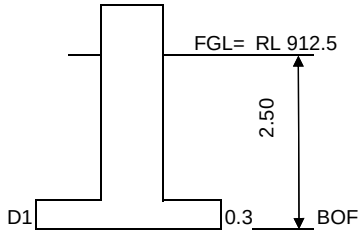
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Foundation MKD. **F1**



PLAN VIEW



SECTIONAL VIEW

Foundation Level Details :

RL of Finish Ground Level ,FGL =	RL	912.5	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		600	kN/m^2

Material Properties :

Unit Weight of Soil =		19.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2	m
Footing dimension along Z direction,B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap =	0.5	x	0.5	=	0.25	m^2
Area of Footing , A =	2.00	x	1.50	=	3	m^2
Zz-z =	2	x	2.00 x 1.50	/ 6	=	1	m^3
Zx-x =	1.5	x	1.5 x 2	/ 6	=	0.75	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	22.5	kN
Self weight of Column =	-1.88	kN
Wt of soil above ftg =	115	kN
Total weight,W=	136	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.5	m
Pedestal dimension along Z direction,Lp2=	0.5	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 19 = 648 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F1 STAAD node103	Force from Load Combinations						Bearing Pressure Calculation					Check for Overturning:									Check for Sliding					
	Axial Force			Moment								About Z - Axis				About X - Axis										
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zz+Mx/Zx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zz-Mx/Zx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2+Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	kN/m^2		kN/m^2		kN-m	kN-m			kN-m	kN-m			kN	kN	kN	kN		
201	-18.5	-271	-2	-5.9	0	55.6	154.0	647.5	safe	27.0	ok	55.6	271	4.89	ok	5.9	203	34.2	ok	18.5	2.0	18.6	271	54.8	ok
202	-18.5	-271	-2	-5.9	0	55.6	154.0	647.5	safe	27.0	ok	55.6	271	4.89	ok	5.9	203	34.2	ok	18.5	2.0	18.6	271	54.8	ok
203	-10.9	-206	-0.2	-0.5	0	32.7	101.9	647.5	safe	35.3	ok	32.7	206	6.3	ok	0.5	154	308	ok	10.9	0.2	10.9	206		ok
204	3.648	-155	-0.1	-0.4	0	-11	63.2	647.5	safe	40.3	ok	10.9	155	14.2	ok	0.4	116	305	ok	3.6	0.1	3.7	155	489	ok
205	-4.74	-195	-5.2	-16	0	14.2	100.1	647.5	safe	30.1	ok	14.2	195	13.7	ok	15.6	146	9.41	ok	4.7	5.2	7.0	195	15.1	ok
206	-2.5	-166	4.89	14.7	0	7.5	82.3	647.5	safe	28.2	ok	7.5	166	22.1	ok	14.7	124	8.47	ok	2.5	4.9	5.5	166	13.5	ok
207	-6.78	-195	-0.2	-0.5	0	20.4	85.9	647.5	safe	44.0	ok	20.4	195	9.57	ok	0.5	146	316	ok	6.8	0.2	6.8	195	506	ok
208	-0.45	-166	-0.1	-0.4	0	1.36	57.3	647.5	safe	53.5	ok	1.4	166	122	ok	0.4	125	296	ok	0.5	0.1	0.5	166	474	ok
209	-3.67	-187	-2.8	-8.3	0	11	84.6	647.5	safe	40.4	ok	11.0	187	17	ok	8.3	140	16.9	ok	3.7	2.8	4.6	187	27.1	ok
210	-3.57	-174	2.48	7.43	0	10.7	78.5	647.5	safe	37.3	ok	10.7	174	16.2	ok	7.4	130	17.5	ok	3.6	2.5	4.3	174	28.1	ok
211	-16	-267	-1	-3.1	0	47.9	141.2	647.5	safe	37.1	ok	47.9	267	5.58	ok	3.1	200	64.4	ok	16.0	1.0	16.0	267	103	ok
212	-5.08	-229	-1	-3	0	15.2	95.8	647.5	safe	57.3	ok	15.2	229	15.1	ok	3.0	172	56.9	ok	5.1	1.0	5.2	229	91.1	ok
213	-11.4	-259	-4.8	-14	0	34.1	139.8	647.5	safe	33.2	ok	34.1	259	7.62	ok	14.4	195	13.5	ok	11.4	4.8	12.3	259	21.6	ok
214	-9.69	-237	2.76	8.27	0	29.1	119.2	647.5	safe	39.1	ok	29.1	237	8.17	ok	8.3	178	21.5	ok	9.7	2.8	10.1	237	34.4	ok
215	-12.9	-259	-1	-3.1	0	38.7	129.2	647.5	safe	43.6	ok	38.7	259	6.7	ok	3.1	194	63.1	ok	12.9	1.0	12.9	259	101	ok
216	-8.15	-238	-1	-3.1	0	24.5	107.8	647.5	safe	50.7	ok	24.5	238	9.72	ok	3.1	178	58.4	ok	8.2	1.0	8.2	238	93.5	ok
217	-10.6	-254	-3	-9	0	31.7	128.2	647.5	safe	40.9	ok	31.7	254	8.01	ok	9.0	190	21.2	ok	10.6	3.0	11.0	254	33.9	ok
218	-10.5	-243	0.94	2.83	0	31.5	116.4	647.5	safe	45.9	ok	31.5	243	7.73	ok	2.8	182	64.4	ok	10.5	0.9	10.5	243	103	ok
219	-9.44	-188	-0.1	-0.3	0	28.3	91.4	647.5	safe	33.9	ok	28.3	188	6.64	ok	0.3	141	435	ok	9.4	0.1	9.4	188	696	ok
220	5.096	-137	-0.1	-0.2	0	-15	61.4	647.5	safe	30.3	ok	15.3	137	9	ok	0.2	103	500	ok	5.1	0.1	5.1	137	797	ok
221	-3.29	-177	-5.1	-15	0	9.87	89.6	647.5	safe	28.8	ok	9.9	177	18	ok	15.4	133	8.65	ok	3.3	5.1	6.1	177	13.8	ok
222	-1.05	-148	4.95	14.9	0	3.16	72.3	647.5	safe	26.4	ok	3.2	148	46.8	ok	14.9	111	7.47	ok	1.1	5.0	5.1	148	12	ok
223	-5.34	-177	-0.1	-0.3	0	16	75.4	647.5	safe	42.7	ok	16.0	177	11.1	ok	0.3	133	466	ok	5.3	0.1	5.3	177	745	ok
224	0.993	-148	-0.1	-0.2	0	-3	52.8	647.5	safe	46.2	ok	3.0	148	49.8	ok	0.2	111	456	ok	1.0	0.1	1.0	148	732	ok
225	-2.22	-170	-2.7	-8.1	0	6.66	74.1	647.5	safe	39.1	ok	6.7	170	25.5	ok	8.1	127	15.6	ok	2.2	2.7	3.5	170	25	ok
226	-2.12	-156	2.53	7.6	0	6.37	68.5	647.5	safe	35.5	ok	6.4	156	24.5	ok	7.6	117	15.4	ok	2.1	2.5	3.3	156	24.6	ok

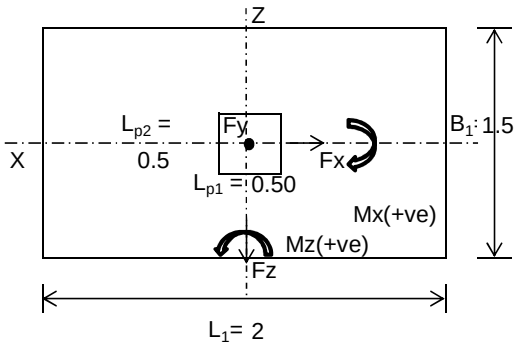
Design pressure for Strength Calculation

Foundation Mkd. F1 STAAD node103	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment			P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
	Fx	Fy	Fz	Mx	My	Mz				
L/C	kN	kN	kN	kN-m	KN-m	kN-m	kN/m^2			
101	-5.07	-62	-0.2	-0.6	0	15.2	36.8	6.4	35.1	4.7
102	-20.7	-216	-2.2	-6.6	0	62.2	143.0	18.7	125.5	1.2
103	-28.7	-217	-3.1	-9.4	0	86.1	170.8	-1.5	145.8	-26.4
104	-10.2	-73	-0.2	-0.6	0	30.5	55.8	-5.3	54.2	-6.8
105	1.471	-33	-0.2	-0.5	0	-4.4	16.1	7.3	14.9	6
106	-5.24	-65	-4.2	-13	0	15.7	54.3	22.9	20.6	-10.9
107	-3.45	-41	3.86	11.6	0	10.3	39.6	19	8.8	-12
108	-20.6	-139	-0.8	-2.4	0	61.7	111.2	-12.3	104.9	-18.6
109	2.68	-58	-0.7	-2.2	0	-8	30.4	14.3	24.6	8.5
110	-10.7	-122	-8.8	-26	0	32.2	108.3	43.9	37.7	-26.8
111	-7.16	-75	7.31	21.9	0	21.5	75.7	32.8	17.3	-25.7
112	-13.5	-119	-0.8	-2.3	0	40.4	83.2	2.4	77.1	-3.8
113	-4.43	-78	-0.8	-2.3	0	13.3	42.4	15.8	36.4	9.8
114	-13.6	-146	-5	-15	0	40.8	109.6	28	69.5	-12.2
115	-8.88	-89	2.99	8.96	0	26.6	68.2	14.9	44.3	-9
116	-14.9	-80	-0.2	-0.5	0	44.7	72.1	-17.3	70.8	-18.6
117	8.37	0	-0.1	-0.3	0	-25	25.4	-24.9	24.6	-25.7
118	-5.05	-64	-8.2	-25	0	15.1	69.2	38.9	3.6	-26.8
119	-1.47	-16	7.93	23.8	0	4.4	41.6	32.8	-21.9	-30.7
120	-7.78	-60	-0.1	-0.4	0	23.3	44.1	-2.7	43.0	-3.8
121	1.264	-19	-0.1	-0.4	0	-3.8	10.8	3.2	9.8	2.3
122	-3.33	-50	-3.9	-12	0	9.98	42.1	22.2	11.1	-8.9
123	-3.19	-30	3.61	10.8	0	9.56	34.1	15	5.2	-14

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PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	12	191.1	150	12	209.5	200

Footing Dimension :

Footing dimension along X direction, L_1	=	2	m
Footing dimension along Z direction, B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Column Dimension :

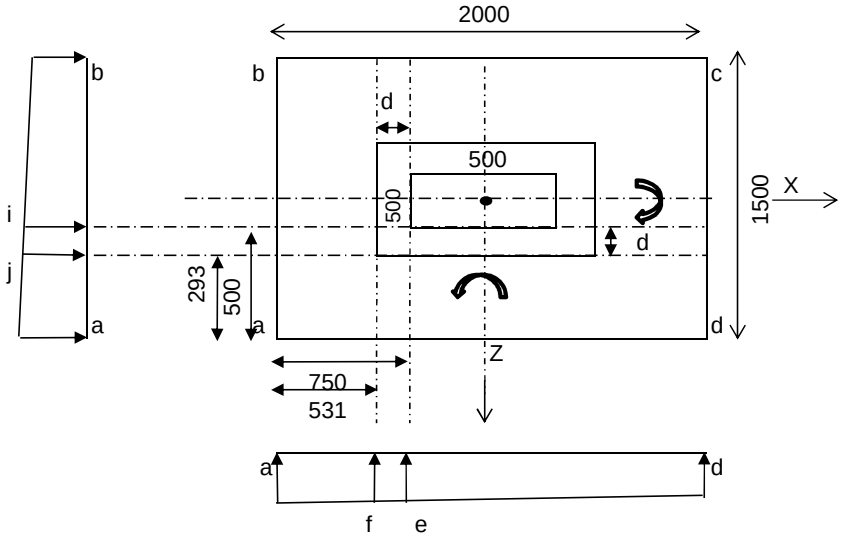
Pedestal dimension along X direction,Lp1	=	0.5	m
Pedestal dimension along Z direction,Lp2	=	0.5	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 170.8 kN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*170.8]*1.5*0.75*0.75	=	72.1	kN-m
Shear force at Face "f"	= 170.8*0.531*1.5	=	136.0	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*170.8]*2*0.5*0.5	=	42.7	kN-m
Shear force at Face "j"	= 170.8*0.293*2	=	100.1	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 2000 mm)

Cantilever Length		=	750	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	219	mm
Bending Moment at "e"		M_u =	72.1	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0094	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y (d-a/2)$	0.9*420*(219-0.0094As/2)As=72.1*10^6	A_s =	887.9	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*1500*300	$A_{s,min}$ =	810	mm^2
Spacing required	1500*22/28*12^2/887.9	S_{reqd} =	191.1	mm c/c
Spacing Provided		S_{prov} =	150	mm c/c
A_s Provided	1500*22/28*12^2/150	$A_{s,pro}$ =	1131.5	mm^2
a = $A_s f_y$/0.85f'$_c$b	1131.5*420/(0.85*35*1500)	a =	10.65	mm
c = a/β_1	10.65/0.8	c =	13.32	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/13.32*(219-13.32)	ε_t =	0.046	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	136.0	kN
Moment at "f"	= [0.5*170.8]*1.5*0.531*0.531	M_u =	36.1	kN-m
ρ_w provided	1131.5/1500/219	$\rho_{w,pro}$ =	0.0034	%
Effective Shear Area	=219*1500/10^6	bd =	0.329	m^2

V_c = lesser of

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$ =	326.8	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	330.2	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	563.6	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	1207.1	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 1500 mm)

Cantilever Length		=	500	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	207	mm
Bending Moment at "i"		M_u =	42.7	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0071	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(207-0.0071As/2)As=42.7*10^6	A_s =	550.9	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*2000*300	$A_{s,min}$ =	1080	mm^2
Spacing required	2000*22/28*12^2/1080	S_{reqd} =	209.52	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2000*22/28*12^2/200	$A_{s,pro}$ =	1131.5	mm^2
a = $A_{st} f_y$/0.85f'$_c$b	1131.5*420/(0.85*35*2000)	a =	7.99	mm
c = a/β_1	7.99/0.8	c =	9.99	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/9.99*(207-9.99)	ε_t =	0.059	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	100.1	kN
Moment at "j"	= [0.5*170.8]*2*0.293*0.293	M_u =	14.7	kN-m
ρ_w provided	= 1131.5/2000/207	$\rho_{w,pro}$ =	0.0027	%
Effective Shear Area	=207*2000/10^6	bd =	0.414	m^2
V_c = lesser of				
a) (0.16λ$\sqrt{f_c'}$ + 17$\rho_w V_u d/M_u$)bd		$V_{c,a}$ =	419.0	kN
b) (0.16λ$\sqrt{f_c'}$ + 17ρ_w)bd		$V_{c,b}$ =	411.1	kN
c) 0.29λ$\sqrt{f_c'}$bd		$V_{c,c}$ =	710.3	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	1520.7	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	170.80	kN/mm^2
Shear force,V_u	= 170.8*[2*1.5-(0.5+0.219)*(0.5+0.207)]	=	425.58	kN
Shear peremeter,b_0	= 2*[(0.5+0.219)+(0.5+0.207)]	=	2.85	m

V_c =0.17(1+2/β)$\sqrt{f_c'}b_0d$ = 0.51 $\sqrt{f_c'}b_0d$

$V_c = 0.083(\alpha_s d/b_0 + 2)\sqrt{f_c'}b_0d$ = 0.36 $\sqrt{f_c'}b_0d$

For Edge column	α_s	30

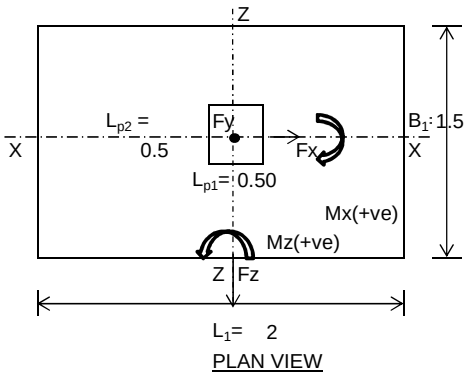
V_c = 0.33$\sqrt{f_c'}b_0d$ = 0.33 $\sqrt{f_c'}b_0d$

φV_c	= 0.75*0.33*SQRT(35)*2852*219/1000	=	914.5	kN
Remarks : if $\varphi V_c > V_u$ ok	= safe			

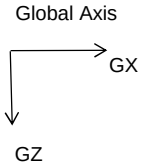
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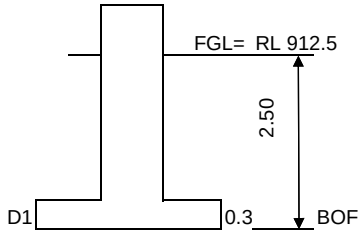
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Foundation MKD. **F1**



PLAN VIEW





SECTIONAL VIEW

Foundation Level Details :

RL of Finish Ground Level ,FGL =	RL	912.5	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		600	kN/m^2

Material Properties :

Unit Weight of Soil =		19.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2	m
Footing dimension along Z direction,B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap =	0.5	x	0.5	=	0.25	m^2
Area of Footing , A =	2.00	x	1.50	=	3	m^2
Zz-z =	2	x	2.00 x 1.50	/ 6	=	1	m^3
Zx-x =	1.5	x	1.5 x 2	/ 6	=	0.75	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	22.5	kN
Self weight of Column =	-1.88	kN
Wt of soil above ftg =	115	kN
Total weight,W=	136	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.5	m
Pedestal dimension along Z direction,Lp2=	0.5	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 19 = 648 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F1 STAAD node104	Force from Load Combinations						Bearing Pressure Calculation					Check for Overturning:								Check for Sliding					
	Axial Force			Moment								About Z - Axis				About X - Axis									
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zz+Mx/Zx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zz-Mx/Zx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2+Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	kN/m^2		kN/m^2		kN-m	kN-m			kN-m	kN-m			kN	kN	kN	kN		
201	-25.3	-250	1.18	3.53	0	76	164.1	647.5	safe	2.7	ok	76.0	250	3.29	ok	3.5	188	53.1	ok	25.3	1.2	25.4	250	85	ok
202	-25.3	-250	1.18	3.53	0	76	164.1	647.5	safe	2.7	ok	76.0	250	3.29	ok	3.5	188	53.1	ok	25.3	1.2	25.4	250	85	ok
203	-10.5	-180	0.11	0.32	0	31.5	91.9	647.5	safe	28.0	ok	31.5	180	5.71	ok	0.3	135	416	ok	10.5	0.1	10.5	180		ok
204	6.933	-158	0.07	0.21	0	-21	73.6	647.5	safe	31.5	ok	20.8	158	7.58	ok	0.2	118	574	ok	6.9	0.1	6.9	158	913	ok
205	0.392	-156	-4.6	-14	0	-1.2	71.8	647.5	safe	32.6	ok	1.2	156	133	ok	13.8	117	8.49	ok	0.4	4.6	4.6	156	13.6	ok
206	-3.96	-181	4.79	14.4	0	11.9	91.4	647.5	safe	29.3	ok	11.9	181	15.2	ok	14.4	136	9.45	ok	4.0	4.8	6.2	181	15.1	ok
207	-5.84	-174	0.1	0.29	0	17.5	75.9	647.5	safe	40.1	ok	17.5	174	9.94	ok	0.3	130	458	ok	5.8	0.1	5.8	174	732	ok
208	2.265	-163	0.08	0.24	0	-6.8	61.6	647.5	safe	47.4	ok	6.8	163	24.1	ok	0.2	123	502	ok	2.3	0.1	2.3	163	807	ok
209	-1.07	-160	-2.5	-7.5	0	3.21	66.6	647.5	safe	40.1	ok	3.2	160	49.8	ok	7.5	120	15.9	ok	1.1	2.5	2.7	160	25.5	ok
210	-2.5	-177	2.69	8.07	0	7.5	77.4	647.5	safe	40.9	ok	7.5	177	23.7	ok	8.1	133	16.5	ok	2.5	2.7	3.7	177	26.4	ok
211	-13.9	-236	0.42	1.26	0	41.7	122.2	647.5	safe	35.4	ok	41.7	236	5.67	ok	1.3	177	141	ok	13.9	0.4	13.9	236	225	ok
212	-0.83	-220	0.39	1.17	0	2.49	77.3	647.5	safe	69.2	ok	2.5	220	88.3	ok	1.2	165	141	ok	0.8	0.4	0.9	220	225	ok
213	-5.73	-219	-3.1	-9.4	0	17.2	102.6	647.5	safe	43.3	ok	17.2	219	12.7	ok	9.4	164	17.5	ok	5.7	3.1	6.5	219	28.1	ok
214	-9	-237	3.93	11.8	0	27	121.7	647.5	safe	36.3	ok	27.0	237	8.78	ok	11.8	178	15.1	ok	9.0	3.9	9.8	237	24.1	ok
215	-10.4	-232	0.41	1.23	0	31.2	110.2	647.5	safe	44.5	ok	31.2	232	7.43	ok	1.2	174	141	ok	10.4	0.4	10.4	232	226	ok
216	-4.33	-224	0.4	1.2	0	13	89.3	647.5	safe	60.1	ok	13.0	224	17.2	ok	1.2	168	140	ok	4.3	0.4	4.3	224	224	ok
217	-6.83	-221	-1.5	-4.6	0	20.5	100.5	647.5	safe	47.1	ok	20.5	221	10.8	ok	4.6	166	35.8	ok	6.8	1.5	7.0	221	57.2	ok
218	-7.9	-234	2.36	7.07	0	23.7	111.3	647.5	safe	45.0	ok	23.7	234	9.89	ok	7.1	176	24.9	ok	7.9	2.4	8.2	234	39.8	ok
219	-9.79	-167	0.07	0.22	0	29.4	85.3	647.5	safe	26.0	ok	29.4	167	5.68	ok	0.2	125	573	ok	9.8	0.1	9.8	167	913	ok
220	7.647	-144	0.03	0.1	0	-23	71.3	647.5	safe	25.1	ok	22.9	144	6.3	ok	0.1	108	1084	ok	7.6	0.0	7.6	144	1751	ok
221	1.106	-143	-4.6	-14	0	-3.3	69.7	647.5	safe	26.0	ok	3.3	143	43.2	ok	13.9	108	7.72	ok	1.1	4.6	4.8	143	12.4	ok
222	-3.25	-168	4.75	14.3	0	9.74	84.7	647.5	safe	27.2	ok	9.7	168	17.2	ok	14.3	126	8.83	ok	3.2	4.8	5.8	168	14.1	ok
223	-5.12	-161	0.06	0.18	0	15.4	69.3	647.5	safe	38.1	ok	15.4	161	10.5	ok	0.2	121	674	ok	5.1	0.1	5.1	161	1072	ok
224	2.979	-150	0.05	0.14	0	-8.9	59.3	647.5	safe	41.0	ok	8.9	150	16.8	ok	0.1	113	817	ok	3.0	0.0	3.0	150	1307	ok
225	-0.36	-147	-2.5	-7.6	0	1.07	60.3	647.5	safe	37.7	ok	1.1	147	137	ok	7.6	110	14.4	ok	0.4	2.5	2.6	147	23.1	ok
226	-1.79	-164	2.66	7.96	0	5.35	70.8	647.5	safe	38.9	ok	5.4	164	30.7	ok	8.0	123	15.5	ok	1.8	2.7	3.2	164	24.8	ok

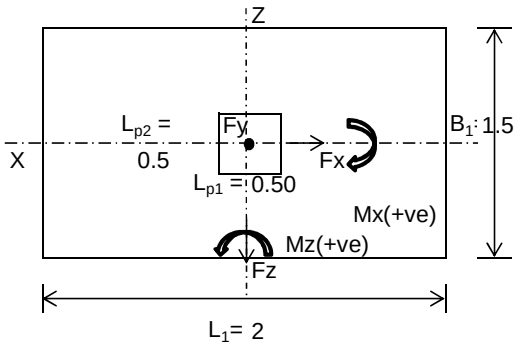
Design pressure for Strength Calculation

Foundation Mkd. F1 STAAD node104	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment						
	Fx	Fy	Fz	Mx	My	Mz	P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
L/C	kN	kN	kN	kN-m	KN-m	kN-m	kN/m^2			
101	-2.5	-46	0.12	0.37	0	7.5	23.3	8.3	22.3	7.3
102	-15.2	-180	0.84	2.52	0	45.5	108.8	17.9	102.1	11.2
103	-37.7	-183	1.77	5.32	0	113	181.4	-45	167.2	-59.2
104	-9.12	-48	0.12	0.37	0	27.3	43.9	-10.9	42.9	-11.9
105	4.833	-30	0.09	0.27	0	-14	25.0	-4.1	24.3	-4.8
106	-0.4	-29	-3.7	-11	0	1.2	25.7	23.3	-3.6	-6
107	-3.88	-49	3.86	11.6	0	11.7	43.5	20.2	12.6	-10.8
108	-19.8	-96	0.35	1.04	0	59.4	93.0	-26	90.2	-28.8
109	8.086	-61	0.29	0.86	0	-24	45.7	-2.9	43.5	-5.2
110	-2.38	-59	-7.2	-22	0	7.14	55.7	41.4	-2.0	-16.3
111	-9.35	-98	7.83	23.5	0	28	92.1	36	29.5	-26.7
112	-11.6	-86	0.33	0.98	0	34.9	65.0	-5	62.4	-7.6
113	-0.08	-71	0.31	0.92	0	0.23	25.2	24.8	22.8	22.3
114	-6.56	-70	-3.3	-9.8	0	19.7	56.1	16.8	29.9	-9.5
115	-6.88	-91	4.03	12.1	0	20.7	67.2	25.9	34.9	-6.5
116	-15.6	-47	0.11	0.33	0	46.7	62.9	-30.6	62.0	-31.4
117	12.34	-12	0.05	0.14	0	-37	41.1	-33	40.8	-33.4
118	1.877	-10	-7.4	-22	0	-5.6	38.7	27.5	-20.9	-32.1
119	-5.09	-49	7.6	22.8	0	15.3	62.0	31.5	1.2	-29.4
120	-7.39	-37	0.09	0.27	0	22.2	34.9	-9.6	34.2	-10.3
121	4.18	-22	0.07	0.21	0	-13	20.2	-5	19.6	-5.6
122	-0.59	-17	-3.6	-11	0	1.76	22.0	18.5	-7.2	-10.7
123	-2.63	-42	3.8	11.4	0	7.88	37.1	21.3	6.7	-9.2

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PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	12	179.9	150	12	209.5	200

Footing Dimension :

Footing dimension along X direction, L_1	=	2	m
Footing dimension along Z direction, B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Column Dimension :

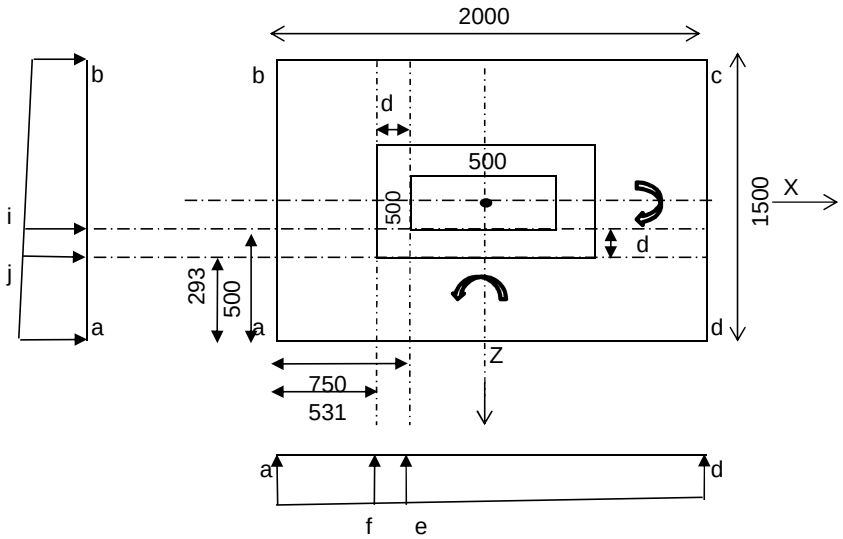
Pedestal dimension along X direction,Lp1	=	0.5	m
Pedestal dimension along Z direction,Lp2	=	0.5	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 181.4 kN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*181.4]*1.5*0.75*0.75	=	76.5	kN-m
Shear force at Face "f"	= 181.4*0.531*1.5	=	144.5	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*181.4]*2*0.5*0.5	=	45.4	kN-m
Shear force at Face "j"	= 181.4*0.293*2	=	106.3	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 2000 mm)

Cantilever Length			=	750	mm
Dia of Bar		d_b	=	12	mm
Effective Depth		d	=	219	mm
Bending Moment at "e"		M_u	=	76.5	kN-m
$C = 0.85f_c'ba$ and $T = A_sf_y$	Therefore 'a' = $f_y/0.85f_c'b * A_s$	a	=	0.0094	$*A_s$
$\varphi M_n \geq M_u = \varphi A_sf_y(d-a/2)$	0.9*420*(219-0.0094As/2)As=76.5*10^6	A_s	=	943.2	mm^2
P min	[SBC 304 Table 7.6.1.1]		=	0.18	%
$A_{s,min}$ =	0.18/100*1500*300	$A_{s,min}$	=	810	mm^2
Spacing required	1500*22/28*12^2/943.2	S_{reqd}	=	179.9	mm c/c
Spacing Provided		S_{prov}	=	150	mm c/c
A_s Provided	1500*22/28*12^2/150	$A_{s,pro}$	=	1131.5	mm^2
$a = A_sf_y/0.85f'_cb$	1131.5*420/(0.85*35*1500)	a	=	10.65	mm
$c = a/\beta_1$	10.65/0.8	c	=	13.32	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/13.32*(219-13.32)	ε_t	=	0.046	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u	=	144.5	kN
Moment at "f"	= [0.5*181.4]*1.5*0.531*0.531	M_u	=	38.4	kN-m
ρ_w provided	1131.5/1500/219	$\rho_{w,pro}$	=	0.0034	%
Effective Shear Area	=219*1500/10^6	bd	=	0.329	m^2

V_c = lesser of

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$	=	326.8	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$	=	330.2	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$	=	563.6	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$			=	1207.1	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok				ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$	=	Not required	

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In Z Direction (Along Length 1500 mm)

Cantilever Length		=	500	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	207	mm
Bending Moment at "i"		M_u =	45.4	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0071	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(207-0.0071As/2)As=45.35*10^6	A_s =	585.5	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*2000*300	$A_{s,min}$ =	1080	mm^2
Spacing required	2000*22/28*12^2/1080	S_{reqd} =	209.52	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2000*22/28*12^2/200	$A_{s,pro}$ =	1131.5	mm^2
a = $A_{st} f_y$/0.85f'$_c$b	1131.5*420/(0.85*35*2000)	a =	7.99	mm
c = a/β_1	7.99/0.8	c =	9.99	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/9.99*(207-9.99)	ε_t =	0.059	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	106.3	kN
Moment at "j"	= [0.5*181.4]*2*0.293*0.293	M_u =	15.6	kN-m
ρ_w provided	= 1131.5/2000/207	$\rho_{w,pro}$ =	0.0027	%
Effective Shear Area	=207*2000/10^6	bd =	0.414	m^2
V_c = lesser of				
a) (0.16λ$\sqrt{f_c'}$ + 17$\rho_w V_u d/M_u$)bd		$V_{c,a}$ =	419.0	kN
b) (0.16λ$\sqrt{f_c'}$ + 17ρ_w)bd		$V_{c,b}$ =	411.1	kN
c) 0.29λ$\sqrt{f_c'}$bd		$V_{c,c}$ =	710.3	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	1520.7	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	181.40	kN/mm^2
Shear force,V_u	= 181.4*[2*1.5-(0.5+0.219)*(0.5+0.207)]	=	451.99	kN
Shear peremeter,b_0	= 2*[(0.5+0.219)+(0.5+0.207)]	=	2.85	m

V_c =0.17(1+2/β)$\sqrt{f_c'}b_0d$ = 0.51 $\sqrt{f_c'}b_0d$

$V_c = 0.083(\alpha_s d/b_0 + 2)\sqrt{f_c'}b_0d$ = 0.36 $\sqrt{f_c'}b_0d$ — For Edge column, α_s = 30

V_c = 0.33$\sqrt{f_c'}b_0d$ = 0.33 $\sqrt{f_c'}b_0d$

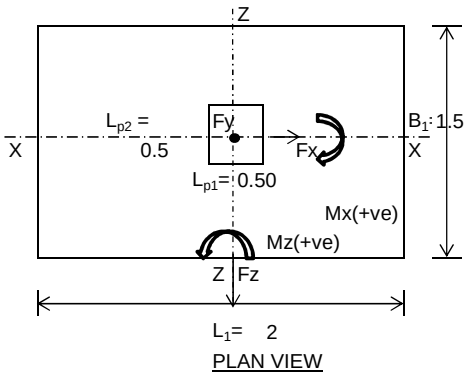
φV_c	= 0.75*0.33*SQRT(35)*2852*219/1000	=	914.5	kN

Remarks : if $\varphi V_c > V_u$ ok = safe

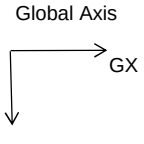
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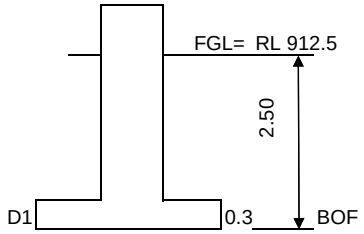
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Foundation MKD. **F1**



PLAN VIEW





SECTIONAL VIEW

Foundation Level Details :

RL of Finish Ground Level ,FGL =	RL	912.5	m
Depth of foundation from FGL. ,H1 =	(-)	2.50	m
Net Safe Bearing Capacity considered =		600	kN/m^2

Material Properties :

Unit Weight of Soil =		19.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2	m
Footing dimension along Z direction,B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap =	0.5	x	0.5	=	0.25	m^2
Area of Footing , A =	2.00	x	1.50	=	3	m^2
Zz-z =	2	x	2.00 x 1.50	/ 6	=	1	m^3
Zx-x =	1.5	x	1.5 x 2	/ 6	=	0.75	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	22.5	kN
Self weight of Column =	-1.88	kN
Wt of soil above ftg =	115	kN
Total weight,W=	136	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.5	m
Pedestal dimension along Z direction,Lp2=	0.5	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 2.5 x 19 = 648 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F1 STAAD node105	Force from Load Combinations						Bearing Pressure Calculation					Check for Overturning:									Check for Sliding					
	Axial Force			Moment								About Z - Axis				About X - Axis										
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zz+Mx/Zxx	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zz-Mx/Zxx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2+Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	kN/m^2		kN/m^2		kN-m	kN-m			kN-m	kN-m			kN	kN	kN	kN		
201	-0.06	-274	-0.5	-1.6	0	0.18	93.7	647.5	safe	89.1	ok	0.2	274	1548	ok	1.6	205	129	ok	0.1	0.5	0.5	274	207	ok
202	-0.06	-274	-0.5	-1.6	0	0.18	93.7	647.5	safe	89.1	ok	0.2	274	1548	ok	1.6	205	129	ok	0.1	0.5	0.5	274	207	ok
203	-0.43	-168	-0.1	-0.2	0	1.29	57.6	647.5	safe	54.4	ok	1.3	168	130	ok	0.2	126	611	ok	0.4	0.1	0.4	168		ok
204	0.431	-168	-0.1	-0.3	0	-1.3	57.6	647.5	safe	54.3	ok	1.3	168	130	ok	0.3	126	475	ok	0.4	0.1	0.4	168	763	ok
205	0	-178	-4.7	-14	0	0	78.4	647.5	safe	40.5	ok	0.0	178	-	-	14.2	134	9.42	ok	0.0	4.7	4.7	178	15.1	ok
206	0	-157	4.58	13.7	0	0	70.8	647.5	safe	34.2	ok	0.0	157	-	-	13.7	118	8.59	ok	0.0	4.6	4.6	157	13.8	ok
207	0	-168	-0.1	-0.2	0	0	56.3	647.5	safe	55.8	ok	0.0	168	-	-	0.2	126	586	ok	0.0	0.1	0.1	168	933	ok
208	0	-168	-0.1	-0.3	0	0	56.3	647.5	safe	55.6	ok	0.0	168	-	-	0.3	126	491	ok	0.0	0.1	0.1	168	789	ok
209	0	-177	-2.7	-8	0	0	69.6	647.5	safe	48.3	ok	0.0	177	-	-	8.0	132	16.6	ok	0.0	2.7	2.7	177	26.6	ok
210	0	-159	2.5	7.51	0	0	63.1	647.5	safe	43.1	ok	0.0	159	-	-	7.5	119	15.9	ok	0.0	2.5	2.5	159	25.4	ok
211	-0.32	-249	-0.2	-0.6	0	0.95	84.9	647.5	safe	81.3	ok	0.9	249	263	ok	0.6	187	295	ok	0.3	0.2	0.4	249	472	ok
212	0.331	-249	-0.2	-0.7	0	-1	85.0	647.5	safe	81.2	ok	1.0	249	251	ok	0.7	187	276	ok	0.3	0.2	0.4	249	441	ok
213	0.007	-257	-3.7	-11	0	-0	100.6	647.5	safe	70.8	ok	0.0	257	11682	ok	11.1	193	17.3	ok	0.0	3.7	3.7	257	27.7	ok
214	0.007	-241	3.28	9.83	0	-0	93.6	647.5	safe	67.4	ok	0.0	241	10967	ok	9.8	181	18.4	ok	0.0	3.3	3.3	241	29.5	ok
215	0.007	-249	-0.2	-0.6	0	-0	84.0	647.5	safe	82.3	ok	0.0	249	11329	ok	0.6	187	293	ok	0.0	0.2	0.2	249	468	ok
216	0.007	-249	-0.2	-0.7	0	-0	84.0	647.5	safe	82.1	ok	0.0	249	11320	ok	0.7	187	279	ok	0.0	0.2	0.2	249	447	ok
217	0.007	-256	-2.2	-6.5	0	-0	93.9	647.5	safe	76.6	ok	0.0	256	11623	ok	6.5	192	29.7	ok	0.0	2.2	2.2	256	47.5	ok
218	0.007	-243	1.72	5.15	0	-0	87.8	647.5	safe	74.0	ok	0.0	243	11026	ok	5.2	182	35.3	ok	0.0	1.7	1.7	243	56.5	ok
219	-0.43	-155	-0	-0.1	0	1.29	53.2	647.5	safe	50.3	ok	1.3	155	120	ok	0.1	116	1039	ok	0.4	0.0	0.4	155	1677	ok
220	0.431	-155	-0.1	-0.2	0	-1.3	53.3	647.5	safe	50.2	ok	1.3	155	120	ok	0.2	116	680	ok	0.4	0.1	0.4	155	1088	ok
221	0	-166	-4.7	-14	0	0	74.1	647.5	safe	36.4	ok	0.0	166	-	-	14.1	124	8.8	ok	0.0	4.7	4.7	166	14.1	ok
222	0	-145	4.61	13.8	0	0	66.7	647.5	safe	29.8	ok	0.0	145	-	-	13.8	108	7.85	ok	0.0	4.6	4.6	145	12.6	ok
223	0	-155	-0	-0.1	0	0	52.0	647.5	safe	51.6	ok	0.0	155	-	-	0.1	116	962	ok	0.0	0.0	0.0	155	1552	ok
224	0	-155	-0.1	-0.2	0	0	51.9	647.5	safe	51.5	ok	0.0	155	-	-	0.2	116	717	ok	0.0	0.1	0.1	155	1148	ok
225	0	-164	-2.6	-7.9	0	0	65.2	647.5	safe	44.2	ok	0.0	164	-	-	7.9	123	15.6	ok	0.0	2.6	2.6	164	25	ok
226	0	-146	2.53	7.6	0	0	59.0	647.5	safe	38.7	ok	0.0	146	-	-	7.6	110	14.4	ok	0.0	2.5	2.5	146	23.1	ok

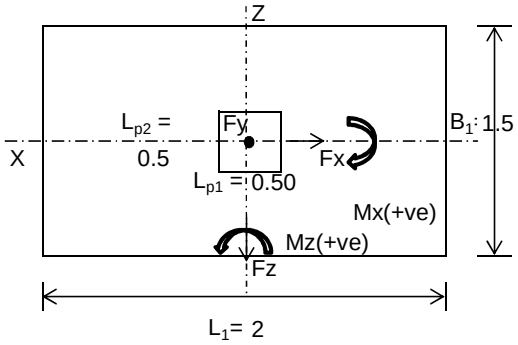
Design pressure for Strength Calculation

Foundation Mkd. F1 STAAD node105	Force from Load Combinations						Design pressure for Strength Calculation			
	Axial Force			Moment			P/A+Mz/Zzz+Mx/Zxx	P/A-Mz/Zzz+Mx/Zxx	P/A+Mz/Zzz-Mx/Zxx	P/A-Mz/Zzz-Mx/Zxx
	Fx	Fy	Fz	Mx	My	Mz				
L/C	kN	kN	kN	kN-m	KN-m	kN-m	kN/m^2			
101	0	-45	-0.1	-0.3	0	0	15.3	15.3	14.5	14.5
102	0.017	-229	-0.4	-1.3	0	-0.1	78.1	78	74.7	74.6
103	-0.08	-226	-0.8	-2.4	0	0.24	78.7	78.2	72.3	71.8
104	-0.35	-38	-0.1	-0.3	0	1.04	14.2	12.1	13.5	11.4
105	0.345	-38	-0.1	-0.3	0	-1	14.2	12.2	13.4	11.3
106	0	-47	-3.8	-11	0	0	30.9	30.9	0.3	0.3
107	0	-30	3.63	10.9	0	0	24.5	24.5	-4.6	-4.6
108	-0.69	-92	-0.2	-0.5	0	2.06	33.6	29.5	32.2	28.1
109	0.695	-92	-0.2	-0.6	0	-2.1	33.7	29.6	32.1	27.9
110	0.005	-109	-7.6	-23	0	-0	67.0	67	5.9	5.9
111	0.005	-76	7.27	21.8	0	-0	54.3	54.3	-3.9	-3.9
112	0.005	-93	-0.2	-0.5	0	-0	31.6	31.6	30.2	30.2
113	0.005	-92	-0.2	-0.6	0	-0	31.6	31.6	30.0	30
114	-0.03	-95	-4	-12	0	0.09	48.1	48	15.8	15.6
115	0.005	-80	3.5	10.5	0	-0	40.7	40.7	12.7	12.7
116	-0.69	-29	-0.1	-0.2	0	2.07	11.9	7.8	11.5	7.3
117	0.69	-29	-0.1	-0.3	0	-2.1	12.0	7.9	11.3	7.2
118	0	-45	-7.5	-23	0	0	45.3	45.3	-15.0	-15
119	0	-12	7.38	22.1	0	0	33.5	33.5	-25.6	-25.6
120	0	-29	-0.1	-0.2	0	0	9.9	9.9	9.4	9.4
121	0	-28	-0.1	-0.2	0	0	9.9	9.9	9.2	9.2
122	0	-41	-3.8	-11	0	0	28.8	28.8	-1.4	-1.4
123	0	-16	3.62	10.9	0	0	19.9	19.9	-9.1	-9.1

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PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	12	209.5	150	12	209.5	200

Footing Dimension :

Footing dimension along X direction, L_1	=	2	m
Footing dimension along Z direction, B_1	=	1.5	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Column Dimension :

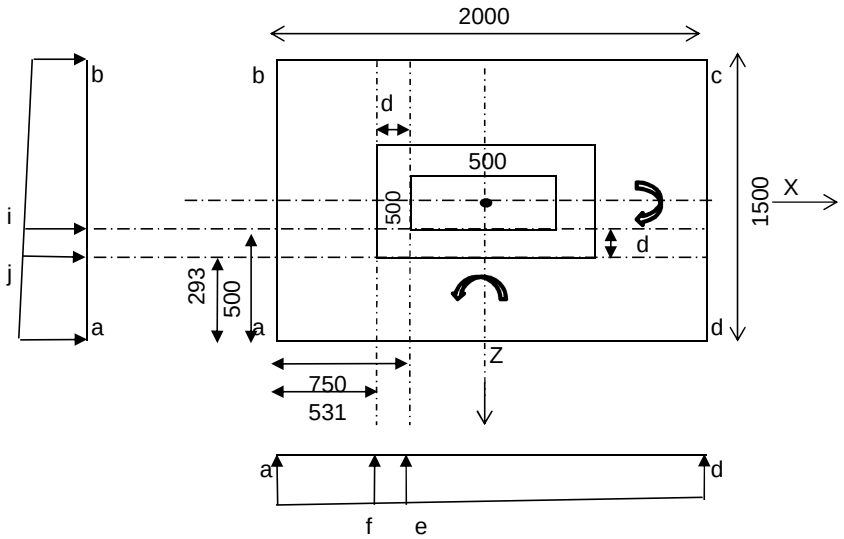
Pedestal dimension along X direction, Lp1	=	0.5	m
Pedestal dimension along Z direction, Lp2	=	0.5	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Design Base Pressure : 78.7 kN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*78.7]*1.5*0.75*0.75	=	33.2	kN-m
Shear force at Face "f"	= 78.7*0.531*1.5	=	62.7	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*78.7]*2*0.5*0.5	=	19.7	kN-m
Shear force at Face "j"	= 78.7*0.293*2	=	46.1	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 2000 mm)

Cantilever Length		=	750	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	219	mm
Bending Moment at "e"		M_u =	33.2	kN-m
C = $0.85f_c'ba$ and T = A_sf_y	Therefore 'a' = $f_y/0.85f_c'b * A_s$	a =	0.0094	$*A_s$
$\varphi M_n \geq M_u = \varphi A_sf_y(d-a/2)$	0.9*420*(219-0.0094As/2)As=33.2*10^6	A_s =	404.6	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*1500*300	$A_{s,min}$ =	810	mm^2
Spacing required	1500*22/28*12^2/810	S_{reqd} =	209.5	mm c/c
Spacing Provided		S_{prov} =	150	mm c/c
A_s Provided	1500*22/28*12^2/150	$A_{s,pro}$ =	1131.5	mm^2
$a = A_sf_y/0.85f'_cb$	1131.5*420/(0.85*35*1500)	a =	10.65	mm
$c = a/\beta_1$	10.65/0.8	c =	13.32	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/13.32*(219-13.32)	ε_t =	0.046	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	62.7	kN
Moment at "f"	= [0.5*78.7]*1.5*0.531*0.531	M_u =	16.6	kN-m
ρ_w provided	1131.5/1500/219	$\rho_{w,pro}$ =	0.0034	%
Effective Shear Area	=219*1500/10^6	bd =	0.329	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$ =	326.8	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	330.2	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	563.6	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	1207.1	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 1500 mm)

Cantilever Length		=	500	mm
Dia of Bar		d_b =	12	mm
Effective Depth		d =	207	mm
Bending Moment at "i"		M_u =	19.7	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0071	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(207-0.0071As/2)As=19.68*10^6	A_s =	252.6	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*2000*300	$A_{s,min}$ =	1080	mm^2
Spacing required	2000*22/28*12^2/1080	S_{reqd} =	209.52	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2000*22/28*12^2/200	$A_{s,pro}$ =	1131.5	mm^2
a = $A_{st} f_y$/0.85f'$_c$b	1131.5*420/(0.85*35*2000)	a =	7.99	mm
c = a/β_1	7.99/0.8	c =	9.99	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/9.99*(207-9.99)	ε_t =	0.059	

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	46.1	kN
Moment at "j"	= [0.5*78.7]*2*0.293*0.293	M_u =	6.8	kN-m
ρ_w provided	= 1131.5/2000/207	$\rho_{w,pro}$ =	0.0027	%
Effective Shear Area	=207*2000/10^6	bd =	0.414	m^2
V_c = lesser of				
a) (0.16λ$\sqrt{f_c'}$ + 17$\rho_w V_u d/M_u$)bd		$V_{c,a}$ =	419.0	kN
b) (0.16λ$\sqrt{f_c'}$ + 17ρ_w)bd		$V_{c,b}$ =	411.1	kN
c) 0.29λ$\sqrt{f_c'}$bd		$V_{c,c}$ =	710.3	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	1520.7	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	78.70	kN/mm^2
Shear force,V_u	= 78.7*[2*1.5-(0.5+0.219)*(0.5+0.207)]	=	196.10	kN
Shear peremeter,b_0	= 2*[(0.5+0.219)+(0.5+0.207)]	=	2.85	m

V_c =0.17(1+2/β)$\sqrt{f_c'}b_0d$ = 0.51 $\sqrt{f_c'}b_0d$

V_c = 0.083(α_sd/b_0 + 2)$\sqrt{f_c'}b_0d$ = 0.36 $\sqrt{f_c'}b_0d$ — For Edge column, α_s = 30

V_c = 0.33$\sqrt{f_c'}b_0d$ = 0.33 $\sqrt{f_c'}b_0d$

φV_c	= 0.75*0.33*SQRT(35)*2852*219/1000	=	914.5	kN

Remarks : if $\varphi V_c > V_u$ ok = safe

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Foundation Base Pressure for F2 :

	Node	L/C	Horizontal Fx KN/m2	Vertical Fy KN/m2	Horizontal Fz KN/m2
Max Px	1	101	0	0	0
Min Px	1	101	0	0	0
Max Py	324	101	0	51.98	0
Min Py	1	101	0	0	0
Max Pz	1	101	0	0	0
Min Pz	1	101	0	0	0

Gross Maximum Base pressure, p_{max} = 52.0 KN/m^2

Net Allowable Safe Bearing Capacity = 600 KN/m^2
Depth of soil above footing = 2.5 m
Unit weight of soil = 19 KN/m^3

Gross Safe Bearing Capacity = 647.5 KN/m^2
> 52.0 KN/m^2 safe

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Design of Foundation F2:

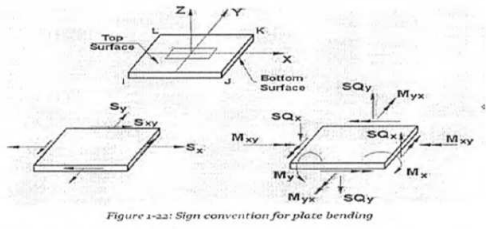
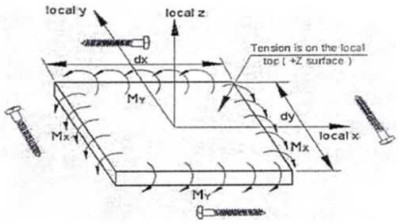


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G1 : F_2_MOMENT At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I			
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C	
Max Qx	124	0.27	-0.025	0	0	0	-13.78	-8.402	5.346	-19.13	-13.75	201 1.4DL	
Min Qx	172	-0.282	-0.271	0	0	0	13.07	35.24	0	13.07	35.24	201 1.4DL	
Max Qy	44	-0.036	0.272	0	0	0	-37.54	-0.339	-6.369	-43.9	-6.708	201 1.4DL	
Min Qy	171	0.062	-0.302	0	0	0	-8.481	43.29	8.797	-17.28	52.09	201 1.4DL	
Max Sx	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Min Sx	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Max Sy	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Min Sy	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Max Sxy	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Min Sxy	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Max Mx	156	-0.203	0.142	0	0	0	16.18	5.977	9.682	25.86	15.66	201 1.4DL	
Min Mx	180	0.193	-0.09	0	0	0	-62.07	-2.109	-2.936	-65.01	-5.045	201 1.4DL	
Max My	62	0.055	-0.242	0	0	0	-6.323	44.95	3.809	-10.13	48.76	201 1.4DL	
Min My	122	0.058	-0.279	0	0	0	-30.59	-26.23	-6.03	-36.62	-32.26	201 1.4DL	
Max Mxy	172	-0.282	-0.271	0	0	0	13.07	35.24	20.31	33.38	55.55	201 1.4DL	
Min Mxy	12	-0.245	-0.23	0	0	0	-45.55	-15.46	-17.98	-63.53	-33.44	201 1.4DL	

PLATE CENTER STRESS ; G2 : F_2_SHEAR At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I			
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C	
Max Qx	124	0.27	-0.025	0	0	0	-13.78	-8.402	5.346	-19.13	-13.75	201 1.4DL	
Min Qx	172	-0.282	-0.271	0	0	0	13.07	35.24	0	13.07	35.24	201 1.4DL	
Max Qy	44	-0.036	0.272	0	0	0	-37.54	-0.339	-6.369	-43.9	-6.708	201 1.4DL	
Min Qy	171	0.062	-0.302	0	0	0	-8.481	43.29	8.797	-17.28	52.09	201 1.4DL	
Max Sx	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Min Sx	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Max Sy	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Min Sy	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Max Sxy	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Min Sxy	292	0.028	0.02	0	0	0	-0.622	0.577	0.341	-0.963	0.918	201 1.4DL	
Max Mx	156	-0.203	0.142	0	0	0	16.18	5.977	9.682	25.86	15.66	201 1.4DL	
Min Mx	180	0.193	-0.09	0	0	0	-62.07	-2.109	-2.936	-65.01	-5.045	201 1.4DL	
Max My	62	0.055	-0.242	0	0	0	-6.323	44.95	3.809	-10.13	48.76	201 1.4DL	
Min My	122	0.058	-0.279	0	0	0	-30.59	-26.23	-6.03	-36.62	-32.26	201 1.4DL	
Max Mxy	172	-0.282	-0.271	0	0	0	13.07	35.24	20.31	33.38	55.55	201 1.4DL	
Min Mxy	12	-0.245	-0.23	0	0	0	-45.55	-15.46	-17.98	-63.53	-33.44	201 1.4DL	

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

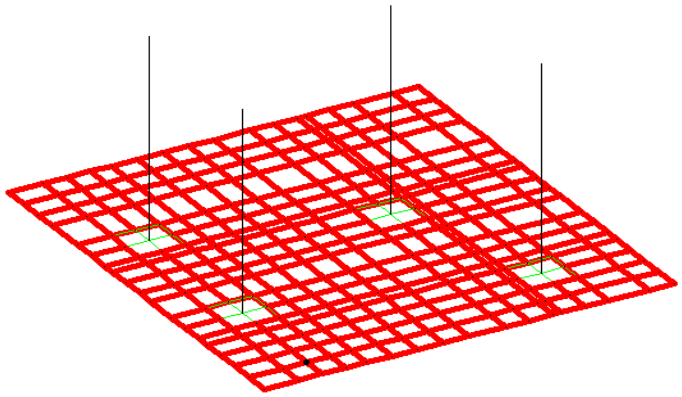
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c'bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	25.86	top face	50	342	25.86		200.9		16	279.37	200	1005.8	14.18	17.73	0.055	tension control
Min Mx	45.00	top face	50	342	45	0.014	350.6	720	16	279.37	200	1005.8	14.18	17.73	0.055	tension control
Max Mxy	33.38	top face	50	342	33.38		259.6		16	279.37	200	1005.8	14.18	17.73	0.055	tension control
Min Mxy	-63.53	bottom face	75	317	63.53	*As	536.5		16	279.37	200	1005.8	14.18	17.73	0.051	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.27		108	19.13	0.003	0.342	356.8	340.8	fc'587	340.83	1257	> 108	ok	255.6	φVc>Vu	not required
	Min Qx	0.282		112.8	13.07			374.2			340.83	1257	> 112.8	ok	255.6	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c'bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	48.76	top face	50	344	48.76		377.9		12	299.38	200	565.8	7.978	9.972	0.1	tension control
Min My	-32.26	bottom face	75	319	32.26	0.014	269.1	720	12	314.29	200	565.8	7.978	9.972	0.093	tension control
Max Mxy	55.55	top face	50	344	55.55		431.0		12	262.49	200	565.8	7.978	9.972	0.1	tension control
Min Mxy	-33.44	bottom face	75	319	33.44	*As	279.0		12	314.29	200	565.8	7.978	9.972	0.093	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.272		108.8	6.708	0.002	0.344	418.6	340.8	fc'587	340.83	356	> 108.8	ok	255.6	φVc>Vu	not required
	Min Qy	0.302		120.8	52.09			337.3			337.29	353	> 120.8	ok	253	φVc>Vu	not required



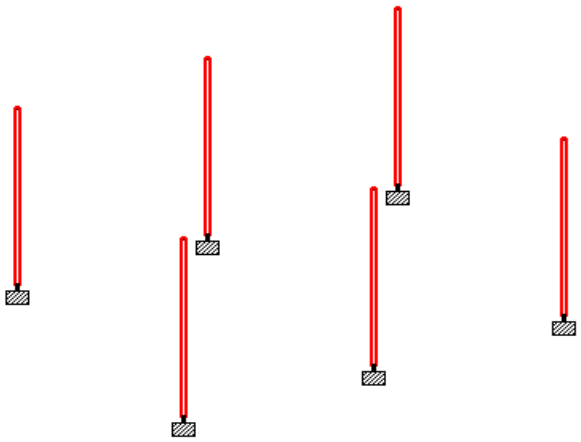
Moment & Shear taken at the face of the column

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7.2. Design of Columns

Design of column C1



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Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.90

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.72

Y_b = (b-d')/b = 0.72

Unsupported length of column parallel to X-X axis, l_{ux} = 2.7 m kl_{ux}/r_b = 18

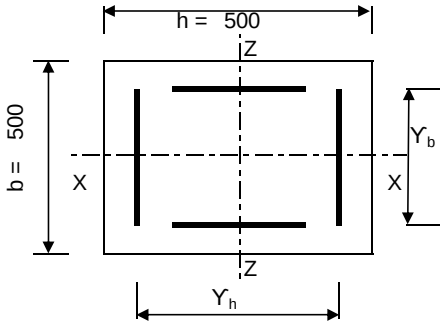
Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.7 m kl_{uz}/r_h = 18

< 22 neglect slenderness effect

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6735.18 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
Max Fx	229	1.3	0.1	254	1.4	0.1	0.029	0.060	262.5	0.01	0.060	262.5	0.00		0.001	safe
Max My	122.2	26.5	32.2	136	29.4	35.8	0.016	0.060	262.5	0.11	0.060	262.5	0.14		0.07	safe
Min My	114.0	26.4	17.9	127	29.3	19.9	0.014	0.060	262.5	0.11	0.060	262.5	0.08	1.61	0.05	safe
Max Mz	183.4	5.3	113.1	204	5.9	125.7	0.023	0.060	262.5	0.02	0.060	262.5	0.48		0.309	safe
Min Mz	205.1	6.4	110.8	228	7.2	123.1	0.026	0.060	262.5	0.03	0.060	262.5	0.47		0.299	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	8	20	+	0	0	2515	1.01	10	320	250



P_n : Nominal axial load

M_n : Nominal moment

P_u : Factored design axial load

M_u : Factored design moment

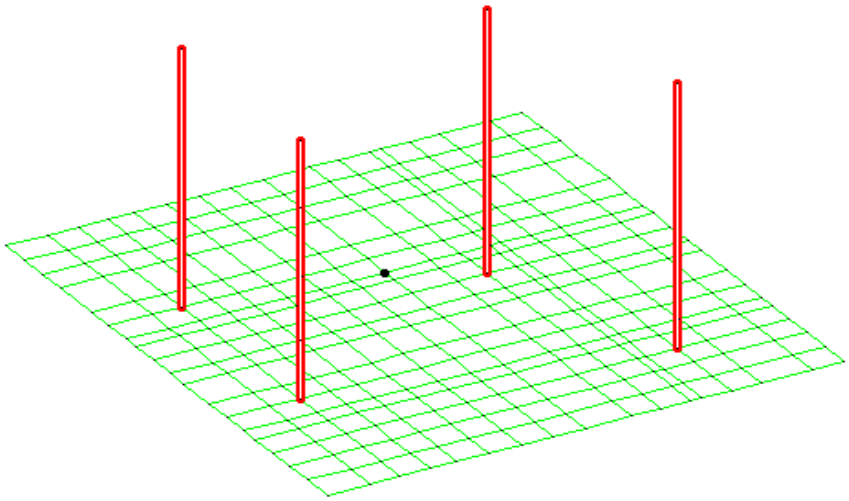
Use interaction diagram R35-420.7

ACI DESIGN HANDBOOK-SP-17M(09)

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Design of column C2



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Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.90

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 450 mm

Depth parallel to Z-Z axis, b = 450 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.69

Y_b = (b-d')/b = 0.69

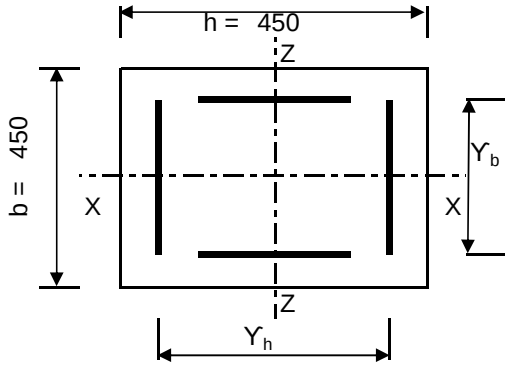
Unsupported length of column parallel to X-X axis, l_{ux} = 2.7 m kl_{ux}/r_b = 20 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.7 m kl_{uz}/r_h = 20

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 5604.68 KN safe

L/C	Factored Force Value P_u	M_{ux}	M_{uz}	Design force Value $P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis $R_{nx} = M_{nx}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	about ZZ axis $R_{nz} = M_{nz}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	Check for Interaction Ratio $\alpha=\log0.5/\log\beta$	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
Max Fx	19	44.7	99.8	21	49.6	110.9	0.003	0.060	191	0.26	0.060	191.4	0.58	1.61	0.530	safe
Max My	0.0	44.7	0.0	0	49.6	0.0	0	0.060	191	0.26	0.060	191.4	0.00		0.11	safe
Min My	19.0	44.7	99.8	21	49.6	110.9	0.003	0.060	191	0.26	0.060	191.4	0.58		0.53	safe
Min Mz	19.0	44.7	99.8	21	49.6	110.9	0.003	0.060	191	0.26	0.060	191.4	0.58		0.530	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db				A_{st} provided(mm^2)	Reinforcement provided (%)		Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2025	8	20	+	0	0	2515	1.24	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.7

ACI DESIGN HANDBOOK-SP-17M(09)

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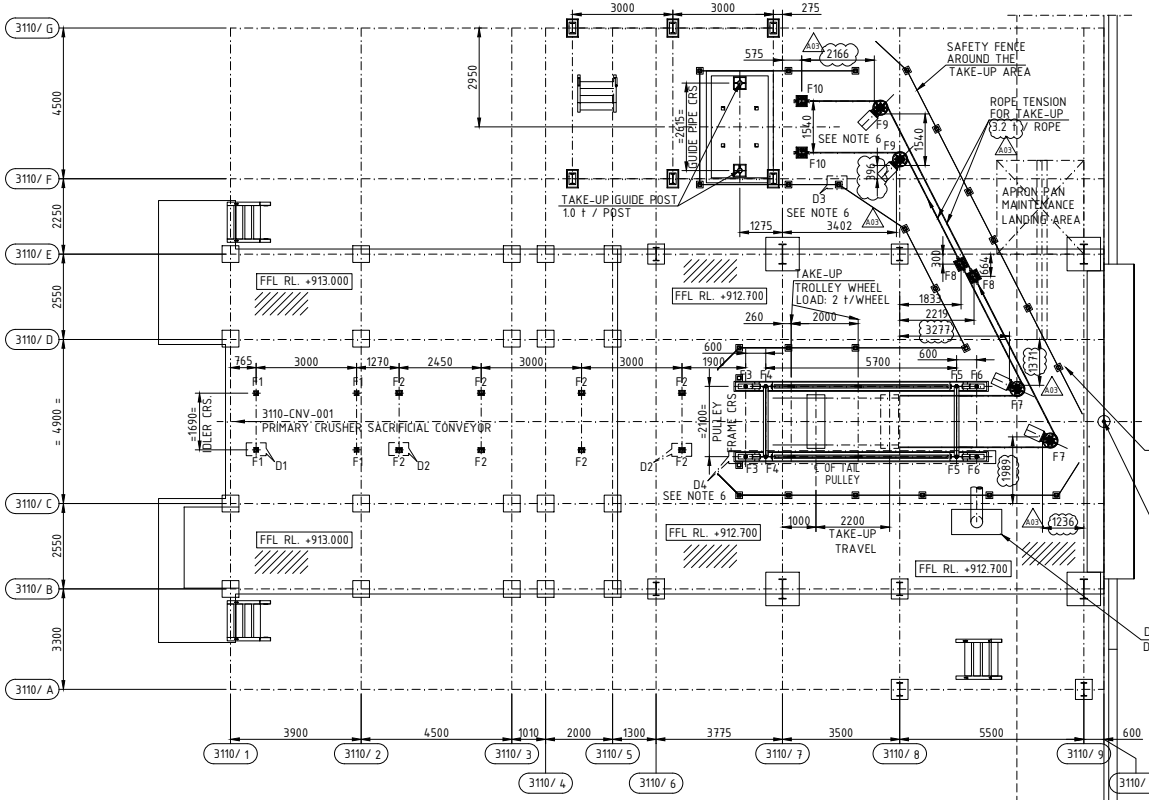
ATTACHMENT 1 MECHANICAL INPUT (GA OF PRIMARY CRUSHER HOUSE)

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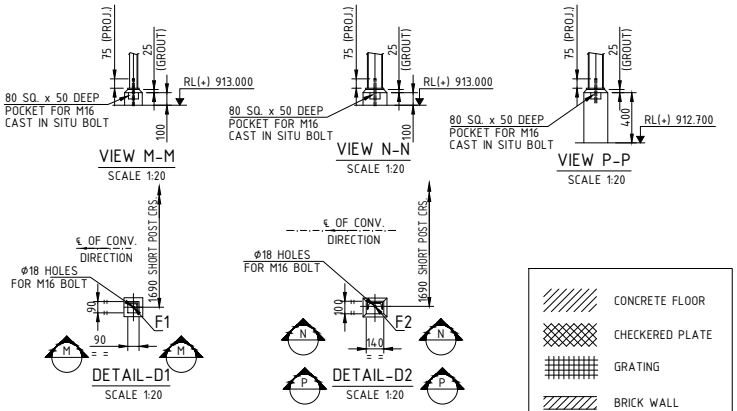
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DRAWING NUMBER MD-216-3110-EG-ME-LPX-1002 REV A03



PLAN AT FFL RL. +913.000
SCALE 1:75



CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.
6. FOR DETAIL D3, D4, F7, F8, F9, F10 REEFER DRAWING 1008

KEY PLAN

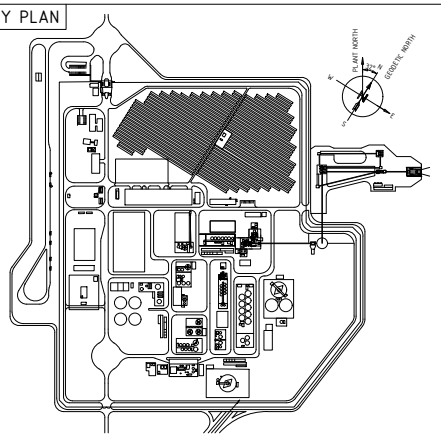


TABLE-3 : LOAD TABLE FOR SHORT POST

FOR CONV. 3110-CNV-001

	POINT	ELEVATION IN "m"	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
			DEAD LOAD	LIVE LOAD	ALONG	ACROSS
NORMAL ZONE	F1	(+)913.000	(+)0.65	(+)0.35	(±)0.11	(±)0.06
IMPACT ZONE	F2	(+)912.500 (+)913.000	(+)0.98	(+)0.55	(±)0.17	(±)0.09

TABLE-4 : LOAD TABLE FOR BUFFER POST

FOR CONV. 3110-CNV-001

POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t")	
			ALONG THE CONV.	ACROSS THE CONV.
F3	(+)912.500	(+)5.6	(+)3.6	(±)0.56
F4		(-)5.6	(+)1.0	(±)0.56
F5		(-)5.6	(-)1.0	(±)0.56
F6		(+)5.6	(-)3.6	(±)0.56

SIGN CONVENTION
UPWARD (-)
DOWNWARD (+)
TOWARDS HEAD PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design A02

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1009	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1008	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1007	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1009	SECTION	A03	04/04/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER	ANN	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1008	SECTION			COMMENTS DT. 02.04.2020 & REVISED AS MARKED A03					
MD-216-3110-EG-ME-LPX-1001	PLAN VIEW	A02	05/03/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1007	ELEVATION	A01	02/12/2019	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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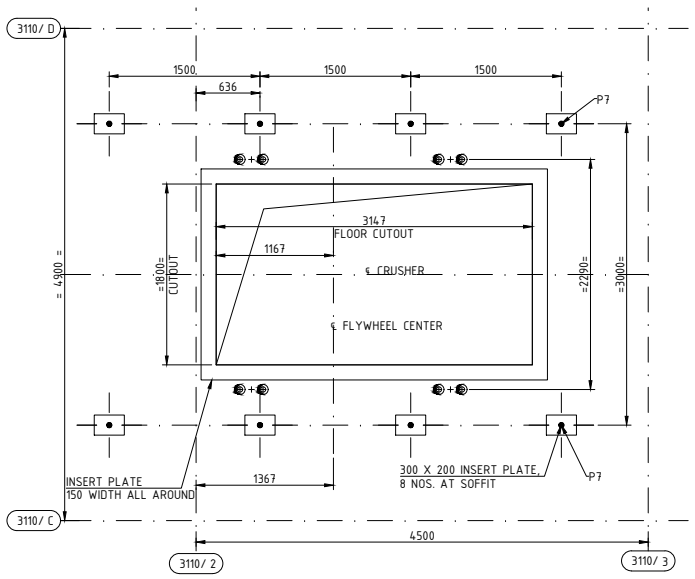
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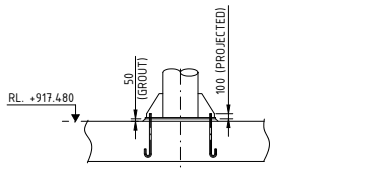
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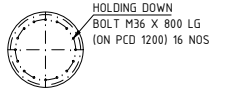
LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_100_H_G_16123_003			2
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31108		OU100666772	2
DRAWN	SKH	29/11/2019	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	30/11/2019	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	02/12/2019	CRUSHING
APPROVED	BGR	02/12/2019	PRIMARY CRUSHER HOUSE - PLAN VIEW
EMPLOYER			
SCALE	1:75	A1	DRAWING NUMBER: MD-216-3110-EG-ME-LPX-1002 REV: A03



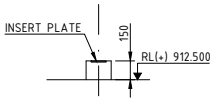
SOFFIT PLAN AT RL. +917.480
SCALE 1:25



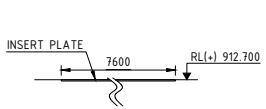
SECTION H-H
SCALE 1:50



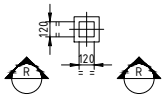
DETAIL D2
SCALE 1:50



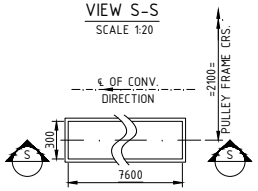
VIEW R-R
SCALE 1:20



VIEW S-S
SCALE 1:20



DETAIL-D3
SCALE 1:20



DETAIL-D4
SCALE 1:20

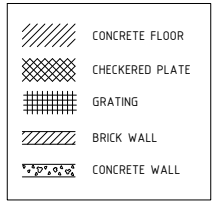
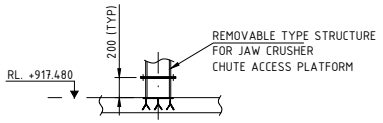


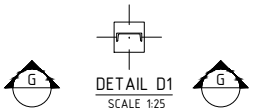
TABLE-8: LOAD FOR CHUTE SUPPORT

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	
		CHUTE EMPTY	CHUTE BLOCKED
P7	(+)917.480 (AT SOFFIT)	(+)2.0	(+)3.5

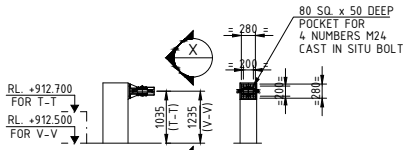
NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L; F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.



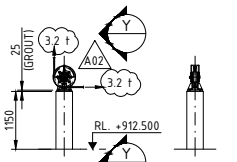
SECTION G-G
SCALE 1:25



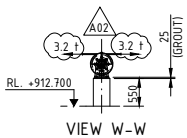
DETAIL D1
SCALE 1:25



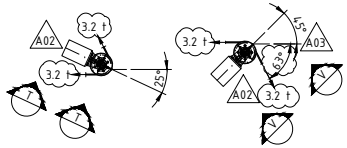
VIEW T-T
VIEW V-V
VIEW X-X
SCALE 1:50



VIEW U-U
VIEW Y-Y
SCALE 1:50

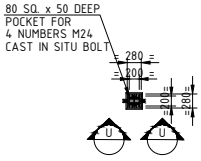


VIEW W-W
SCALE 1:50

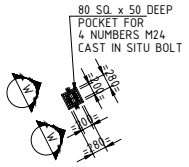


DETAIL-F7
SCALE 1:50

DETAIL-F9
SCALE 1:50

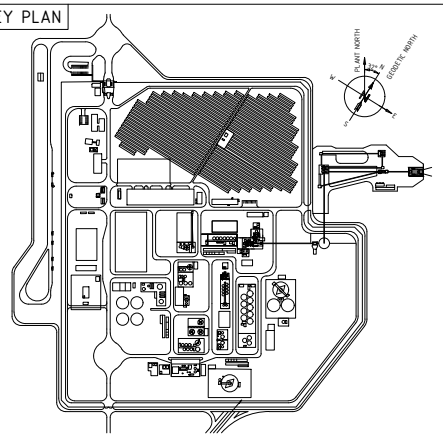


DETAIL-F10
SCALE 1:50



DETAIL-F8
SCALE 1:50

KEY PLAN



REV	DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER	ANN	BRD	BD	BGR	
	COMMENTS DT. 02.04.2020 & REVISED AS MARKED					
A01	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE
MD-216-3110-EG-ME-LPX-1009	PLAN VIEW		
MD-216-3110-EG-ME-LPX-10087	PLAN VIEW		
MD-216-3110-EG-ME-ELV-1008	SECTION		
MD-216-3110-EG-ME-LPX-1002	PLAN VIEW	A02	04/04/2020
MD-216-3110-EG-ME-LPX-1001	PLAN VIEW		
MD-216-3110-EG-ME-ELV-1007	ELEVATION	A01	05/03/2020

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16123_007	REVISION 1
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31157	DOCUMENT ID 0U100672983 / REVISION 1
DRAWN	ANN	02/03/2020
CHECKED	BRD	03/03/2020
ENGINEER	BD	04/03/2020
APPROVED	BGR	05/03/2020
EMPLOYER		
SCALE	1:75, 1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
CRUSHING
PRIMARY CRUSHER HOUSE - PLAN VIEW

DRAWING NUMBER MD-216-3110-EG-ME-LPX-1008 REV A02

ATTACHMENT 2 STAAD OUTPUT OF "MD-216-3110-EG-CV-CAL-1005-A01_HGTU TOWER"

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 14, 2020                *
*           Time=    19:26:54                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\samrat dey 14.10.15\7. Maaden\3.Primary Crusher\DESIGN\grade slab\STAAD\MD-216-3110-EG-.. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 24-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 913 0; 2 3 913 0; 3 6 913 0; 4 0 913 4.5; 5 3 913 4.5; 6 6 913 4.5
     9. 100 0 910 0; 101 0 910 4.5; 102 3 910 0; 103 3 910 4.5; 104 6 910 0
    10. 105 6 910 4.5
    11. MEMBER INCIDENCES
    12. 237 100 1; 238 102 2; 239 104 3; 240 101 4; 241 103 5; 242 105 6
    13. DEFINE MATERIAL START
    14. ISOTROPIC CONCRETE
    15. E 2.21467E+006
    16. POISSON 0.17
    17. DENSITY 2.5
    18. ALPHA 1E-005
    19. DAMP 0.05
    20. TYPE CONCRETE
    21. STRENGTH FCU 2812.28
    22. END DEFINE MATERIAL
    23. **
    24. MEMBER PROPERTY AMERICAN
    25. 237 TO 242 PRIS YD 0.5 ZD 0.5
    26. CONSTANTS
    27. MATERIAL CONCRETE ALL
    28. SUPPORTS
    29. 100 TO 105 FIXED
    30. ***
    31. DEFINE REFERENCE LOADS
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    32. LOAD R1 LOADTYPE DEAD  TITLE DEAD LOAD(DL)
    33. JOINT LOAD
    34. 1 FX 0 FY -1.945 FZ -0.009
    35. 2 FX -0.181 FY -2.686 FZ -0.015
    36. 3 FX 0.182 FY -1.454 FZ -0.009
```

```
37. 4 FX -0.37 FY -2.015 FZ 0.009
38. 5 FX 0.369 FY -2.653 FZ 0.015
39. 6 FX 0 FY -1.371 FZ 0.008
40. SELFWEIGHT Y -1 LIST ALL
41. LOAD R2 LOADTYPE LIVE  TITLE LIVE LOAD(LL)
42. JOINT LOAD
43. 1 FX -0.001 FY -3.135 FZ 0
44. 2 FX -0.549 FY -7.791 FZ -0.104
45. 3 FX 0.554 FY -4.416 FZ -0.034
46. 4 FX -0.945 FY -3.217 FZ 0
47. 5 FX 0.938 FY -8.456 FZ 0.111
48. 6 FX 0.003 FY -4.558 FZ 0.027
49. LOAD R3 LOADTYPE DEAD  TITLE EQUIPMENT LOAD(EQL)
50. JOINT LOAD
51. 1 FX 0 FY 0.483 FZ 0
52. 2 FX -0.194 FY -0.805 FZ 0.009
53. 3 FX 0.205 FY -3.63 FZ -0.009
54. 4 FX -0.008 FY -0.166 FZ 0
55. 5 FX 0.001 FY -0.785 FZ 0.008
56. 6 FX -0.004 FY -6.497 FZ -0.008
57. LOAD R4 LOADTYPE WIND  TITLE WIND LOAD WL(+)X
58. JOINT LOAD
59. 1 FX 0.051 FY 1.347 FZ 0
60. 2 FX 0.592 FY -0.216 FZ 0.001
61. 3 FX 0.889 FY -1.131 FZ -0.002
62. 4 FX 0.742 FY 2.572 FZ 0
63. 5 FX 0.741 FY -2.571 FZ 0.002
64. 6 FX 0.044 FY -0.001 FZ -0.001
65. LOAD R5 LOADTYPE WIND  TITLE WIND LOAD WL(-)X
66. JOINT LOAD
67. 1 FX -0.051 FY -1.347 FZ 0
68. 2 FX -0.592 FY 0.216 FZ -0.001
69. 3 FX -0.889 FY 1.131 FZ 0.002
70. 4 FX -0.742 FY -2.572 FZ 0
71. 5 FX -0.741 FY 2.571 FZ -0.002
72. 6 FX -0.044 FY 0.001 FZ 0.001
73. LOAD R6 LOADTYPE WIND  TITLE WIND LOAD WL+Z
74. JOINT LOAD
75. 1 FX 0 FY 0.552 FZ 0.32
76. 2 FX 0.013 FY 1.159 FZ 0.52
77. 3 FX -0.222 FY 1.242 FZ 0.479
78. 4 FX 0.095 FY -0.38 FZ 0.319
79. 5 FX 0.114 FY -1.504 FZ 0.514
80. 6 FX 0 FY -1.069 FZ 0.475
81. LOAD R7 LOADTYPE WIND  TITLE WIND LOAD WL(-)Z
82. JOINT LOAD
83. 1 FX 0 FY -0.552 FZ -0.32
84. 2 FX -0.013 FY -1.159 FZ -0.52
85. 3 FX 0.222 FY -1.242 FZ -0.479
86. 4 FX -0.095 FY 0.38 FZ -0.319
87. 5 FX -0.114 FY 1.504 FZ -0.514
88. 6 FX 0 FY 1.069 FZ -0.475
89. LOAD R8 LOADTYPE SEISMIC  TITLE SEISMIC LOAD SL(+)X
90. JOINT LOAD
91. 1 FX 0 FY 1.211 FZ 0
92. 2 FX 0.338 FY -0.444 FZ 0.001
```

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 93. 3 FX 0.59 FY -0.766 FZ -0.001
 94. 4 FX 0.462 FY 2.108 FZ 0
 95. 5 FX 0.461 FY -2.088 FZ 0.001
 96. 6 FX 0 FY -0.02 FZ -0.001
 97. LOAD R9 LOADTYPE SEISMIC  TITLE SEISMIC LOAD SL(-)X
 98. JOINT LOAD
 99. 1 FX 0 FY -1.211 FZ 0
100. 2 FX -0.338 FY 0.444 FZ -0.001
101. 3 FX -0.59 FY 0.766 FZ 0.001
102. 4 FX -0.462 FY -2.108 FZ 0
103. 5 FX -0.461 FY 2.088 FZ -0.001
104. 6 FX 0 FY 0.02 FZ 0.001
105. LOAD R10 LOADTYPE SEISMIC  TITLE SEISMIC LOAD SL(+)Z
106. JOINT LOAD
107. 1 FX 0 FY 0.454 FZ 0.164
108. 2 FX 0.104 FY 1.007 FZ 0.386
109. 3 FX -0.104 FY 1.275 FZ 0.379
110. 4 FX -0.008 FY -0.455 FZ 0.163
111. 5 FX 0.007 FY -1.005 FZ 0.382
112. 6 FX 0 FY -1.276 FZ 0.376
113. LOAD R11 LOADTYPE SEISMIC  TITLE SEISMIC LOAD SL(-)Z
114. JOINT LOAD
115. 1 FX 0 FY -0.454 FZ -0.164
116. 2 FX -0.104 FY -1.007 FZ -0.386
117. 3 FX 0.104 FY -1.275 FZ -0.379
118. 4 FX 0.008 FY 0.455 FZ -0.163
119. 5 FX -0.007 FY 1.005 FZ -0.382
120. 6 FX 0 FY 1.276 FZ -0.376
121. LOAD R12 LOADTYPE TEMPERATURE  TITLE TEMPERATURE LOAD(TL)
122. JOINT LOAD
123. 1 FX -0.01 FY 0.194 FZ -0.055
124. 2 FX -1.606 FY 0.062 FZ -0.026
125. 3 FX 1.643 FY -0.256 FZ -0.068
126. 4 FX -0.615 FY -0.212 FZ 0.055
127. 5 FX 0.581 FY -0.025 FZ 0.068
128. 6 FX 0.007 FY 0.237 FZ 0.027
129. END DEFINE REFERENCE LOADS
130. **
131. *FOR STRENGTH DESIGN
132. LOAD 101 1.4(DL)
133. REFERENCE LOAD
134. R1 1.4
135. LOAD 102 1.4(DL)+1.7(LL+EQL)
136. REFERENCE LOAD
137. R1 1.4 R2 1.7 R3 1.7
138. LOAD 103 1.4(DL+TL)+1.7(LL+EQL)
139. REFERENCE LOAD
140. R1 1.4 R12 1.4 R2 1.7 R3 1.7
141. LOAD 104 1.2(DL)+0.8WL(+X)
142. REFERENCE LOAD
143. R1 1.2 R4 0.8
144. LOAD 105 1.2(DL)+0.8WL(-X)
145. REFERENCE LOAD
146. R1 1.2 R5 0.8
147. LOAD 106 1.2(DL)+0.8WL(+Z)
148. REFERENCE LOAD
```

```
149. R1 1.2 R6 0.8
150. LOAD 107 1.2(DL)+0.8WL(-Z)
151. REFERENCE LOAD
152. R1 1.2 R7 0.8
153. LOAD 108 1.2(DL)+1.6WL(+X)+0.5(LL+EQL)
154. REFERENCE LOAD
155. R1 1.2 R4 1.6 R2 0.5 R3 0.5
156. LOAD 109 1.2(DL)+1.6WL(-X)+0.5(LL+EQL)
157. REFERENCE LOAD
158. R1 1.2 R5 1.6 R2 0.5 R3 0.5
159. LOAD 110 1.2(DL)+1.6WL(+Z)+0.5(LL+EQL)
160. REFERENCE LOAD
161. R1 1.2 R6 1.6 R2 0.5 R3 0.5
162. LOAD 111 1.2(DL)+1.6WL(-Z)+0.5(LL+EQL)
163. REFERENCE LOAD
164. R1 1.2 R7 1.6 R2 0.5 R3 0.5
165. LOAD 112 1.2(DL)+SL(+X)+0.5(LL+EQL)
166. REFERENCE LOAD
167. R1 1.2 R8 1.0 R2 0.5 R3 0.5
168. LOAD 113 1.2(DL)+SL(-X)+0.5(LL+EQL)
169. REFERENCE LOAD
170. R1 1.2 R9 1.0 R2 0.5 R3 0.5
171. LOAD 114 1.2(DL)+SL(+Z)+0.5(LL+EQL)
172. REFERENCE LOAD
173. R1 1.2 R10 1.0 R2 0.5 R2 0.5
174. LOAD 115 1.2(DL)+SL(-Z)+0.5(LL+EQL)
175. REFERENCE LOAD
176. R1 1.2 R11 1.0 R2 0.5 R3 0.5
177. LOAD 116 0.9(DL)+1.6WL(+X)
178. REFERENCE LOAD
179. R1 0.9 R4 1.6
180. LOAD 117 0.9(DL)+1.6WL(-X)
181. REFERENCE LOAD
182. R1 0.9 R5 1.6
183. LOAD 118 0.9(DL)+1.6WL(+Z)
184. REFERENCE LOAD
185. R1 0.9 R6 1.6
186. LOAD 119 0.9(DL)+1.6WL(-Z)
187. REFERENCE LOAD
188. R1 0.9 R7 1.6
189. LOAD 120 0.9(DL)+SL(+X)
190. REFERENCE LOAD
191. R1 0.9 R8 1.0
192. LOAD 121 0.9(DL)+SL(-X)
193. REFERENCE LOAD
194. R1 0.9 R9 1.0
195. LOAD 122 0.9(DL)+SL(+Z)
196. REFERENCE LOAD
197. R1 0.9 R10 1.0
198. LOAD 123 0.9(DL)+SL(-Z)
199. REFERENCE LOAD
200. R1 0.9 R11 1.0
201. **
202. *FOR ALLOWABLE STRESS DESIGN
203. LOAD 201 DL+LL+EQL+TL
204. REFERENCE LOAD
```

```
205. R1 1.0 R2 1.0 R3 1.0 R12 1.0
206. LOAD 202 DL+1.0(LL+EQL)+1.0TL
207. REFERENCE LOAD
208. R1 1.0 R2 1.0 R3 1.0 R12 1.0
209. LOAD 203 DL+WL(+X)
210. REFERENCE LOAD
211. R1 1.0 R4 1.0
212. LOAD 204 DL+WL(-X)
213. REFERENCE LOAD
214. R1 1.0 R5 1.0
215. LOAD 205 DL+WL(+Z)
216. REFERENCE LOAD
217. R1 1.0 R6 1.0
218. LOAD 206 DL+WL(-Z)
219. REFERENCE LOAD
220. R1 1.0 R7 1.0
221. LOAD 207 DL+0.7SL(+X)
222. REFERENCE LOAD
223. R1 1.0 R8 0.7
224. LOAD 208 DL+0.7SL(-X)
225. REFERENCE LOAD
226. R1 1.0 R9 0.7
227. LOAD 209 DL+0.7SL(+Z)
228. REFERENCE LOAD
229. R1 1.0 R10 0.7
230. LOAD 210 DL+0.7SL(-Z)
231. REFERENCE LOAD
232. R1 1.0 R11 0.7
233. LOAD 211 DL+0.75WL(+X)+0.75(LL+EQL)
234. REFERENCE LOAD
235. R1 1.0 R4 0.75 R2 0.75 R3 0.75
236. LOAD 212 DL+0.75WL(-X)+0.75(LL+EQL)
237. REFERENCE LOAD
238. R1 1.0 R5 0.75 R2 0.75 R3 0.75
239. LOAD 213 DL+0.75WL(+Z)+0.75(LL+EQL)
240. REFERENCE LOAD
241. R1 1.0 R6 0.75 R2 0.75 R3 0.75
242. LOAD 214 DL+0.75WL(-Z)+0.75(LL+EQL)
243. REFERENCE LOAD
244. R1 1.0 R7 0.75 R2 0.75 R3 0.75
245. LOAD 215 DL+0.525SL(+X)+0.75(LL+EQL)
246. REFERENCE LOAD
247. R1 1.0 R8 0.525 R2 0.75 R3 0.75
248. LOAD 216 DL+0.525SL(-X)+0.75(LL+EQL)
249. REFERENCE LOAD
250. R1 1.0 R9 0.525 R2 0.75 R3 0.75
251. LOAD 217 DL+0.525SL(+Z)+0.75(LL+EQL)
252. REFERENCE LOAD
253. R1 1.0 R10 0.525 R2 0.75 R3 0.75
254. LOAD 218 DL+0.525SL(-Z)+0.75(LL+EQL)
255. REFERENCE LOAD
256. R1 1.0 R11 0.525 R2 0.75 R3 0.75
257. LOAD 219 0.6(DL)+WL(+X)
258. REFERENCE LOAD
259. R1 0.6 R4 1.0
260. LOAD 220 0.6(DL)+WL(-X)
```

```
261. REFERENCE LOAD
262. R1 0.6 R5 1.0
263. LOAD 221 0.6(DL)+WL(+Z)
264. REFERENCE LOAD
265. R1 0.6 R6 1.0
266. LOAD 222 0.6(DL)+WL(-Z)
267. REFERENCE LOAD
268. R1 0.6 R7 1.0
269. LOAD 223 0.6(DL)+0.7SL(+X)
270. REFERENCE LOAD
271. R1 0.6 R8 0.7
272. LOAD 224 0.6(DL)+0.7SL(-X)
273. REFERENCE LOAD
274. R1 0.6 R9 0.7
275. LOAD 225 0.6(DL)+0.7SL(+Z)
276. REFERENCE LOAD
277. R1 0.6 R10 0.7
278. LOAD 226 0.6(DL)+0.7SL(-Z)
279. REFERENCE LOAD
280. R1 0.6 R11 0.7
281. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**12**	**NUMBER OF MEMBERS**	**6**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 8/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 36
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 195398.8 MB

```
282. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 14,2020 TIME= 19:26:59 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

ATTACHMENT 3 STAAD OUTPUT OF "MD-216-3110-EG-CV-CAL-1005-A01_FOUNDATION F2"

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 14, 2020                *
*           Time=    20: 2:22                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\samrat dey 14.10.15\7. Maaden\3.Primary Crusher\DESIGN\grade slab\STAAD\MD-216-3110-EG-.. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 24-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 -1 0.25; 2 3.9 -1 0.25; 3 0 -1 3.95; 4 3.9 -1 3.95; 5 0.32475 -1 3.95
     9. 6 0.32475 -1 3.71275; 7 0 -1 3.71275; 8 0.5495 -1 3.95; 9 0.5495 -1 3.71275
    10. 10 0.77425 -1 3.95; 11 0.77425 -1 3.71275; 12 0.975 -1 3.95
    11. 13 0.975 -1 3.71275; 14 1.21875 -1 3.95; 15 1.21875 -1 3.71275
    12. 16 1.4625 -1 3.95; 17 1.4625 -1 3.71275; 18 1.70625 -1 3.95
    13. 19 1.70625 -1 3.71275; 20 1.95 -1 3.95; 21 1.95 -1 3.71275; 22 2.29575 -1 3.95
    14. 23 2.29575 -1 3.71275; 24 2.5205 -1 3.95; 25 2.5205 -1 3.71275
    15. 26 2.74525 -1 3.95; 27 2.74525 -1 3.71275; 28 2.825 -1 3.95
    16. 29 2.825 -1 3.71275; 30 3.04975 -1 3.95; 31 3.04975 -1 3.71275
    17. 32 3.2745 -1 3.95; 33 3.2745 -1 3.71275; 34 3.65625 -1 3.95
    18. 35 3.65625 -1 3.71275; 36 3.9 -1 3.71275; 37 0.32475 -1 3.4875; 38 0 -1 3.4875
    19. 39 0.5495 -1 3.4875; 40 0.77425 -1 3.4875; 41 0.975 -1 3.4875
    20. 42 1.21875 -1 3.4875; 43 1.4625 -1 3.4875; 44 1.70625 -1 3.4875
    21. 45 1.95 -1 3.4875; 46 2.29575 -1 3.4875; 47 2.5205 -1 3.4875
    22. 48 2.74525 -1 3.4875; 49 2.825 -1 3.4875; 50 3.04975 -1 3.4875
    23. 51 3.2745 -1 3.4875; 52 3.65625 -1 3.4875; 53 3.9 -1 3.4875
    24. 54 0.32475 -1 3.26225; 55 0 -1 3.26225; 56 0.5495 -1 3.26225
    25. 57 0.77425 -1 3.26225; 58 0.975 -1 3.26225; 59 1.21875 -1 3.26225
    26. 60 1.4625 -1 3.26225; 61 1.70625 -1 3.26225; 62 1.95 -1 3.26225
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    29. 69 3.65625 -1 3.26225; 70 3.9 -1 3.26225; 71 0.32475 -1 3.119; 72 0 -1 3.119
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    37. 96 1.95 -1 2.89375; 97 2.29575 -1 2.89375; 98 2.5205 -1 2.89375
    38. 99 2.74525 -1 2.89375; 100 2.825 -1 2.89375; 101 3.04975 -1 2.89375
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110. 327 3.65625 -1 4.2
111. MEMBER INCIDENCES
112. 257 209 291; 258 90 290; 259 50 293; 260 166 292
113. ELEMENT INCIDENCES SHELL
114. 1 3 5 6 7; 2 5 8 9 6; 3 8 10 11 9; 4 10 12 13 11; 5 12 14 15 13; 6 14 16 17 15
115. 7 16 18 19 17; 8 18 20 21 19; 9 20 22 23 21; 10 22 24 25 23; 11 24 26 27 25
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194. 284 319 320 20 18; 285 320 321 22 20; 286 321 322 24 22; 287 322 323 26 24
195. 288 323 324 28 26; 289 324 325 30 28; 290 325 326 32 30; 291 326 327 34 32
196. 292 327 312 4 34
197. START GROUP DEFINITION
198. ELEMENT
199. _MOMENT 1 TO 28 31 TO 44 47 TO 65 68 TO 81 84 TO 137 140 TO 153 156 TO 177 -
200. 180 TO 193 196 TO 256 261 TO 292
201. _SHEAR 1 TO 28 31 TO 44 47 TO 65 68 TO 81 84 TO 137 140 TO 153 156 TO 177 -
202. 180 TO 193 196 TO 256 261 TO 292
203. MEMBER
204. _C2 257 TO 260
205. END GROUP DEFINITION
206. ELEMENT PROPERTY
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207. 1 TO 256 261 TO 292 THICKNESS 0.4
208. DEFINE MATERIAL START
209. ISOTROPIC CONCRETE
210. E 2.21467E+006
211. POISSON 0.17
212. DENSITY 2.5
213. ALPHA 1E-005
214. DAMP 0.05
215. TYPE CONCRETE
216. STRENGTH FCU 2812.28
217. END DEFINE MATERIAL
218. **
219. MEMBER PROPERTY AMERICAN
220. 257 258 PRIS YD 0.45 ZD 0.45
221. MEMBER PROPERTY AMERICAN
222. 259 260 PRIS YD 0.45 ZD 0.55
223. CONSTANTS
224. BETA 135 MEMB 259 260
225. MATERIAL CONCRETE ALL
226. SUPPORTS
227. 1 TO 256 261 TO 292 PLATE MAT DIRECT Y SUBGRADE 7200 PRINT COMPRESSION
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JOINT	INFLUENCE AREA
1	0.039071
2	0.029326
3	0.039559
4	0.029692
5	0.066936
6	0.063536
7	0.037549
8	0.054755
9	0.051973
10	0.051831
11	0.049198
12	0.054146
13	0.051395
14	0.059384
15	0.056367
16	0.059384
17	0.056367
18	0.059384
19	0.056367
20	0.071808
21	0.068161
22	0.069494
23	0.065964
24	0.054755
25	0.051973
26	0.037092
27	0.035208
28	0.037092
29	0.035208
30	0.054755
31	0.051973

32	0.073879
33	0.070127
34	0.076194
35	0.072323
36	0.028184
37	0.061887
38	0.036575
39	0.050625
40	0.047922
41	0.050062
42	0.054905
43	0.054905
44	0.054905
45	0.066392
46	0.064253
47	0.050625
48	0.034294
49	0.034294
50	0.050625
51	0.068307
52	0.070447
53	0.027452
54	0.050623
55	0.029918
56	0.041410
57	0.039199
58	0.040950
59	0.044911
60	0.044911
61	0.044911
62	0.054308
63	0.052557
64	0.041410
65	0.028052
66	0.028052
67	0.041410
68	0.055874
69	0.057624
70	0.022455
71	0.050623
72	0.029918
73	0.041410
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75	0.040950
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85	0.055874
86	0.057624
87	0.022455

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92	0.050062
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95	0.054905
96	0.066392
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99	0.034294
100	0.034294
101	0.050625
102	0.068307
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105	0.077273
106	0.045668
107	0.063211
108	0.059836
109	0.062508
110	0.068555
111	0.068555
112	0.068555
113	0.082898
114	0.080227
115	0.063211
116	0.042820
117	0.042820
118	0.063211
119	0.085289
120	0.087961
121	0.034277
122	0.070679
123	0.041771
124	0.057817
125	0.054730
126	0.057174
127	0.062705
128	0.062705
129	0.062705
130	0.075824
131	0.073381
132	0.057817
133	0.039166
134	0.039166
135	0.057817
136	0.078011
137	0.080455
138	0.031352
139	0.055293
140	0.032678
141	0.045231
142	0.042816
143	0.044728

144	0.049055
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146	0.049055
147	0.059318
148	0.057407
149	0.045231
150	0.030640
151	0.030640
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153	0.061029
154	0.062941
155	0.024527
156	0.061887
157	0.036575
158	0.050625
159	0.047922
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161	0.054905
162	0.054905
163	0.054905
164	0.066392
165	0.064253
166	0.050625
167	0.034294
168	0.034294
169	0.050625
170	0.068307
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172	0.027452
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174	0.028700
175	0.039725
176	0.037604
177	0.039283
178	0.043083
179	0.043083
180	0.043083
181	0.052097
182	0.050418
183	0.039725
184	0.026910
185	0.026910
186	0.039725
187	0.053599
188	0.055279
189	0.021541
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231	0.077695
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245	0.071509
246	0.078427
247	0.078427
248	0.078427
249	0.094836
250	0.091779
251	0.072313
252	0.048986
253	0.048986
254	0.072313
255	0.097571

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257	0.039213
258	0.063536
259	0.037549
260	0.051973
261	0.049198
262	0.051395
263	0.056367
264	0.056367
265	0.056367
266	0.068161
267	0.065964
268	0.051973
269	0.035208
270	0.035208
271	0.051973
272	0.070127
273	0.072323
274	0.028184
275	0.066112
276	0.054080
277	0.051193
278	0.053479
279	0.058652
280	0.058652
281	0.058652
282	0.070924
283	0.068638
284	0.054080
285	0.036635
286	0.036635
287	0.054080
288	0.072970
289	0.075255
294	0.020297
295	0.015234
296	0.034344
297	0.028094
298	0.026594
299	0.027781
300	0.030469
301	0.030469
302	0.030469
303	0.036844
304	0.035656
305	0.028094
306	0.019031
307	0.019031
308	0.028094
309	0.037906
310	0.039094
311	0.020297
312	0.015234
313	0.034344
314	0.028094
315	0.026594

```
         316          0.027781
         317          0.030469
         318          0.030469
         319          0.030469
         320          0.036844
         321          0.035656
         322          0.028094
         323          0.019031
         324          0.019031
         325          0.028094
         326          0.037906
         327          0.039094
      TOTAL          16.379992
  228. DEFINE REFERENCE LOADS
  229. LOAD R1 LOADTYPE DEAD  TITLE DEAD LOAD
  230. SELFWEIGHT Y -1
  231. JOINT LOAD
  232. 290 291 FX 3.2 FY 3.2
  233. 292 293 FX -3.2
  234. 292 293 FX 1.5 FZ 2.9
  235. ELEMENT LOAD
  236. 1 TO 256 261 TO 292 PR GY -1.14
  237. END DEFINE REFERENCE LOADS
  238. LOAD 101 DL
  239. REFERENCE LOAD
  240. R1 1.0
  241. LOAD 201 1.4DL
  242. REFERENCE LOAD
  243. R1 1.4
  244. PERFORM ANALYSIS


          P R O B L E M   S T A T I S T I C S
          -----------------------------------

    NUMBER OF JOINTS        327  NUMBER OF MEMBERS       4
    NUMBER OF PLATES        288  NUMBER OF SOLIDS        0
    NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    323


         SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

    ORIGINAL/FINAL BAND-WIDTH=   323/    28/     87 DOF
  TOTAL      PRIMARY LOAD CASES =     2, TOTAL DEGREES OF FREEDOM =     993
  TOTAL LOAD COMBINATION  CASES =     0  SO FAR.
    SIZE OF STIFFNESS MATRIX =       87 DOUBLE  KILO-WORDS
    REQRD/AVAIL. DISK SPACE  =     14.5/ 195397.8 MB


*** LOAD CASE   101 -- START ITERATION NO.   2
*** LOAD CASE   101 -- START ITERATION NO.   3
*** LOAD CASE   101 -- START ITERATION NO.   4
```

**NOTE-Tension/Compression converged after 4 iterations, Case= 101

*** LOAD CASE 201 -- START ITERATION NO. 2
*** LOAD CASE 201 -- START ITERATION NO. 3
*** LOAD CASE 201 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 201

245. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 14,2020 TIME= 20: 2:26 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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